Beyond Yields: How Climate Change Affects Farmers Through Market Structure*

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Abstract

Climate change hurts farmers by lowering crop yields. In this paper, I show that it also hurts farmers by changing the structure of the markets thay they sell in. To document this, I exploit extreme drought shocks in the water-intensive Indian sugar industry where 8M smallholders sell perishable cane to oligopsonistic sugar mill intermediaries. I assemble a newly digitized 35-year administrative panel on the universe of Indian sugar mills, and a georeferenced farmer-mill transaction panel on 3M cane farmers. Using a staggered event study design, I show that droughts concentrate markets by removing the least productive mills, and trigger long-term productivity growth within survivors through management upgrading. Yet farm-gate cane prices fall as these gains are captured by survivors through increased buyer power. Reduced cane prices account for one half of farmers' long-term income losses - on par with yield losses themselves - pointing to a large role for market structure in shaping farmers' climate burden. To quantify aggregate welfare impacts, I build and estimate an equilibrium model of the Indian sugar sector that embeds a management upgrading choice for sugar mills. Aggregate mill productivity rises by 4.5% through both selection and within-mill upgrading, but long-term farmer welfare reduces by 1.8% as markdowns go up.

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1 Introduction

A large and compelling literature documents that climate change imposes significant costs on farmers by lowering crop yields.[1] Even after accounting for farmer adaptation, global average yields are projected to decline by more than 20% by the end of the century (Hultgren et al., 2025), implying large losses for farmers. These impacts are expected to be felt most acutely in low-income countries where a large share of the population is still reliant on agriculture. Crop yields, however, form only one part of the story when it comes to farmer welfare. What farmers earn also depends on the prices they receive for their harvest, which are typically set in markets comprising a small number of intermediaries such as traders, agribusinesses or processors.[2] A growing literature shows that these intermediaries wield substantial buyer power over farmers by significantly marking down the prices they pay below what a competitive market would deliver (Chatterjee, 2023; Rubens, 2023; Zavala, 2026; Dominguez-Iino, 2026). Prices that farmers receive are, therefore, intrinsically tied to their local market structure.

Crucially, this market structure is not fixed because intermediaries are themselves exposed to climate change, and their response impacts the prices that farmers receive. For example, a severe climate shock can push the weakest intermediaries to exit, handing greater buyer power to the ones that survive.[3] Similarly, a rise or fall in their productivity can shift demand for inputs and move prices in either direction. Changes in local market structure can therefore either cushion farmers' losses or compound them. Yet we know little about how agricultural markets adjust in response to climate events, and how these adjustments impact farmer welfare. Filling this gap has been especially difficult because changes in long-term market structure and prices are rarely observed. This motivates the central question of the paper: How economically significant is climate-induced restructuring of agricultural markets for long-term farmer welfare?

I take this question to the Indian sugar industry, which is the world's second largest, and produces a fifth of all sugar worldwide. In this setting, farmers sell perishable cane to local sugar mill intermediaries that produce sugar and supply downstream consumer markets. Contributing roughly 2% to India's overall GDP, the sugar sector occupies an

[1] Some of the earlier seminal works include Mendelsohn, Nordhaus and Shaw (1994); Schlenker and Roberts (2009); Lobell, Schlenker and Costa-Roberts (2011)

[2] Traders have been documented by Bergquist and Dinerstein (2020) and Chatterjee (2023) in Kenya and India; Dhingra and Tenreyro (2021), Zavala (2026) and Dominguez-Iino (2026) study agribusinesses in six countries across Africa and South America; Avignon and Guigue (2025) and Rubens (2023) look at processors in France and China.

[3] Kochhar, Song and Sun (2024) find that higher oligopsony power in agricultural markets also blunts farmers' own capacity to adapt due to lower input usage in anticipation of lower farm-gate prices.

important position in India's agricultural and industrial landscape, with 8 million smallholder farmers growing cane and 0.5 million workers employed in its 550 sugar mills. However, climate change has emerged as a real threat, in particular, due to cane's extreme water-intensity, which makes the sector highly sensitive to droughts.[4]

Droughts, the focus of this paper, are a direct consequence of meteorological and hydrological conditions such as failures of the summer monsoon, elevated temperatures that raise evapotranspiration, and the depletion of soil moisture, all of which are likely exacerbated in frequency and severity by climate change.[5] The timing of these events is arguably exogenous to local industry characteristics. Many of the most severe episodes are driven by El Niño events, which are also projected to intensify in a warming world.[6] I focus on extreme, 1-in-50-year drought events because they are severe enough to alter market structure through *direct effects* on farmers, via lower cane yields, and *indirect effects* on sugar mills, via disruptions to local supply chains.

Two features make the Indian sugar setting an ideal laboratory for this paper. First, perishability of cane ties farmers to nearby mills and creates localized markets, with the median cane sale at a distance of only 13 km from a mill. This spatial concentration gives mills oligopsony power and allows me to partition cane-growing regions into distinct markets. Second, different markets experience droughts in different years, which gives quasi-experimental variation in where and when the shock lands, and in how it reshapes local market structure. Because these features characterize agricultural supply chains across a wide range of commodities and geographies[7], the paper's findings promise to be relevant for settings beyond Indian sugar.

In this paper, I make four contributions. First, I newly digitize a novel 35-year administrative panel on the universe of all sugar mills in India and supplement it with a georeferenced farmer-mill transaction panel on 3 million cane farmers and a survey of 220 sugar mills. This data allows me to document the structure and evolution of cane markets throughout India on an unprecedented scale. Second, exploiting spatial and temporal variation in extreme drought events, I provide empirical evidence that exposure to droughts causes the least productive sugar mills to exit, and their farmers to reallocate to nearby survivors. These surviving mills reorganize their labour and improve cane quality management, allowing them to observe long-term productivity gains. Strikingly, within-

[4]Mention press release on vulnerability of Indian sugar sector to droughts

[5]Cite papers on causes and increasing frequency of droughts in India

[6]Cite papers on (1) El Niño - India droughts link; (2) El Niño impacts more severe under climate change

[7]Mention other commodities and regions - like coffee in Rwanda, Costa Rica (Macchiavello et al.); maize in Kenya (in Bergquist and Dinerstein); tobacco in China (Rubens); cash crops in Peru (Zavala)

mill cane prices see a decline indicating that internal productivity gains are captured by surviving mills through amplified markdowns. Third, I develop an equilibrium framework of the Indian sugar sector to rationalize the empirical findings. I combine a nested discrete choice system for farmers, who choose among crops and among sugar mills if they grow cane, with oligopsonistic pricing among mills who also make a management upgrading choice. Fourth, I estimate this model using mill-level and farmer-level data to quantify how drought-induced selection and within-mill productivity growth contribute to aggregate productivity gains among sugar mill intermediaries. More importantly, I estimate how changes in market structure triggered by a transient drought shock impact long-term farmer welfare.

To examine how extreme droughts reshape local market structure and impact cane farmers, I assemble several novel datasets that allow me to link sugar mill operations and farm-gate prices with drought exposure. First, I digitize archival records from the Indian Directorate of Sugar, which document comprehensive information on procurement, production, and managerial and technological practices for the universe of Indian sugar mills from 1990 to 2025. Second, through a close partnership with the Sugar Commissionerate in Maharashtra – India's largest sugar producing state – I obtain first-time access to a georeferenced panel on farmer-mill transactions covering all 3 million cane farmers in the state from 2020 to 2025. Third, I conduct a survey of 220 sugar mills in Maharashtra and elicit information on drought risk, farmer-mill relationships and climate adaptation investments. Next, I link these administrative records to each mill's financial data from 1993 to 2023 using common identifiers with the Annual Survey of Industries (ASI). This allows me to identify, for the first time, every plant in the ASI for a large and important industry.[8] Finally, I construct Standardized Precipitation-Evapotranspiration Indices (SPEI) – my primary measure of extreme droughts – for each market-year, using high resolution data from the Drought Atlas of India (Chuphal et al., 2024).

Using this combined data, I document three stylized facts that directly motivate my empirical strategy and findings. First, trade in cane is highly localized, giving rise to spatially segmented markets. Nearly 80% of all trade occurs within 25 km of a mill location and virtually all trade exhausts within 50 km. This is because cane's sugar content deteriorates rapidly after harvest, leaving mills with only a 12–18 hour window to process it. Second, mills have price-setting power over farmers in these local cane markets but behave as price-takers in the sugar market. This is driven by policy wherein the central government tightly controls mill-level sugar releases to stabilize consumer prices. Third,

[8]The ASI does not disclose plant names or any identifier other than an anonymous plant ID.

mills differ enormously in observed productivity, measured by sugar recovery. The most productive mills extract more than twice as much sugar from a given quantity of cane as the least productive, who also earn thinner margins per tonne of cane crushed. This makes the least productive mills most vulnerable to extreme droughts, setting the stage for my empirical strategy and results.

My empirical strategy exploits spatial and temporal variation in extreme drought events which correspond to SPEI values lower than -2. To quantify market- and mill-level adjustments to droughts, I implement an event-study design using the Sun and Abraham (2021) estimator to address concerns about heterogeneous treatment effects in staggered difference-in-differences settings. The empirical results reveal four key findings. First, extreme droughts concentrate markets by triggering the exit of least productive sugar mills and the reallocation of their farmers to nearby survivors. Marginal survivors who run shorter seasons at baseline and are therefore capacity-unconstrained relative to productive survivors absorb this reallocated cane. Second, these marginal survivors that see a rise in scale improve cane quality management practices. This allows them to raise sucrose content in procured cane and observe a 5 percent increase in sugar recovery – the observed measure of productivity – over the subsequent decade. Third, and strikingly, within-mill cane prices do not change for marginal survivors despite recovery gains, while capacity constrained productive survivors see a 5% decline in their cane prices, suggesting amplified oligopsony power in cane markets. A decomposition of aggregate impacts on farmer cane income into a *yield* component and a *price* component reveals that reduced cane prices explain one half of farmers' income losses – on par with yield losses themselves. Together, these results point to a large role for market structure, a previously unexplored channel, in shaping farmers' long-term climate burden.

To interpret these findings, I develop and estimate an equilibrium model of the Indian sugar sector that links sugar mills' demand for cane with upstream farmer supply. On the demand side, I model sugar mills as oligopsonists and embed capacity constraints and a management upgrading choice in their optimization problem. I define their production function using a Leontief technology to account for the fact that cane cannot be substituted with either labour or capital in order to produce sugar. Departing from the common assumption that the conversion rate of intermediate input to output is homogeneous (Rubens, 2023; Avignon and Guigue, 2025) or exogenous (De Loecker and Scott, 2016), I treat sugar recovery – the conversion factor in my setting – as a mill-level choice increasing in cane management upgrading. This structure yields the paper's two central comparative statics that explain the event-study results. First, upgrading benefit

is increasing in cane volume. This complementarity between scale and upgrading explains how a reallocation-induced rise in scale of marginal survivors drives their recovery growth. Second, the benefit also scales with the share of captured value, which is larger when residual supply is less elastic. So, the same restructuring that raises a survivor's scale also raises the share of the upgrading gain it retains.

On the supply side, I model farmer decisions using a nested discrete-choice framework that captures two margins of adjustment. In the outer nest, a farmer chooses which specific agricultural commodity to grow. In the inner nest, a farmer growing cane chooses which mill to sell to, with choice shaped by the offered price, and observed mill characteristics. This nested structure generates an own-price supply elasticity with two margins – cross-mill substitution (stealing farmers from rival mills) and cross-crop substitution (pulling farmers into cane from alternative crops). This elasticity carries the paper's market-power mechanism: when droughts increase market concentration through mill exit and farmer reallocation, a surviving mill's within-cane share rises, its residual supply becomes less elastic, and its markdown widens.

Estimation proceeds in two stages. First, I estimate the production function using a control-function approach with endogenous exit (Olley and Pakes, 1996; Ackerberg, Caves and Frazer, 2015). To address the Doraszelski and Jaumandreu (2021) critique that input demand may shift due to either productivity or markdowns, I use the nested farmer supply model to control for markdown variation with observed prices of sugar and cane, and mill-level market shares, thereby restoring the scalar-unobservable assumption. Next, I estimate the supply elasticities using crop-level and mill-level shares with instruments for prices. As expected, I find that mills are closer substitutes for one another than cane is for competing crops, suggesting substantial crop switching costs. I also find the cross-crop elasticity estimate to be in line with values from the agricultural trade literature.[9]

The estimates reveal that sugar mills hold considerable buyer power over farmers: the average mill paid its farmers 83% of the marginal revenue product of cane. This markdown level is comparable with figures obtained for other agricultural goods in the literature.[10] Using the estimated TFP and cane price markdowns, I examine how drought-induced market restructuring affected both mill productivity and buyer power. I find that droughts increased within-mill TFP by 12 percent on average, compared to a 5 percent

[9] My estimated value of 2.12 sits in between the range of 1.5-2.5 documented in Sotelo (2020); Bergquist and Dinerstein (2020); Costinot, Donaldson and Smith (2016); Dominguez-Iino (2026).

[10] Average inverse markdowns in Avignon and Guigue (2025); Dominguez-Iino (2026); Zavala (2026); Crespi and Sexton (2005) range between 0.66-0.9. My inverse markdown figure of 0.83 falls in the middle of this range.

rise in sugar recovery (equivalently, the marginal product of cane) documented earlier. Droughts also led to increased exertion of oligopsony power in cane markets: markdowns increased on average by 5-6 percent, a decade after the shock.

Next, I use the model to quantify the impact of extreme droughts on aggregate mill productivity and farmer welfare. First, I decompose the productivity gains from drought-induced restructuring into a *between-firm* resource reallocation component and a *within-firm* improvement component. Lower cane supply reduces variable profits of mills, pushing the weakest ones into losses. Surviving mills, who inherit farmers from exiting mills, respond by improving management practices and observe a rise in productivity. Mills whose productivity is too low to survive, even with these improvements, exit the market. Both responses – productivity improvements among surviving mills and the exit of unproductive ones – raise aggregate productivity by 4.5 percent. A decomposition reveals that approximately 90 percent of this increase comes from the exit of low-productivity sugar mills, with the remaining 10 percent stemming from within-firm productivity improvements. I pair this decomposition with a full welfare accounting for both sides of the market, tracing how each channel maps into mill profits and farmer surplus. Mill profits rise by 4.8 percent in the long-term, however cane farmer welfare reduces by 1.8 percent as fewer survivors in more concentrated markets impose wider markdowns on cane prices.

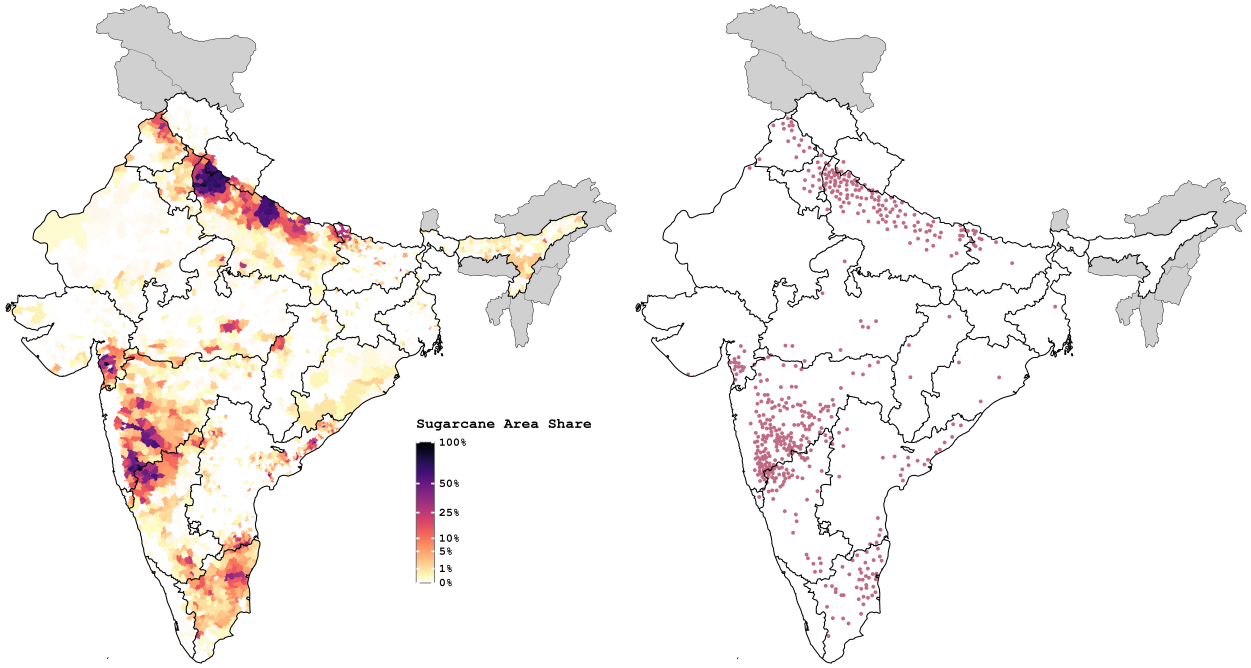
Related Literature: This paper adds to three strands of the literature.

The remainder of the paper is organised as follows. Section 2 provides institutional background on the Indian sugar industry. Section 3 describes the data sources and data collection exercise. Section 4 presents motivating stylized facts. Section 5 outlines the empirical strategy and presents the event-study results. Section 6 develops the structural model. Section 7 discusses identification and estimation. Section 8 presents counterfactual simulations. Section 9 concludes.

2 Institutional Background

India is the world's second-largest producer of sugar behind Brazil, accounting for a fifth of global production. The industry occupies a central position in the Indian economy – it sources cane from roughly 8 million smallholder farmers, supporting 50 million livelihoods, and directly employs half a million industrial workers.[11] In the process, it contributes to approximately 2 percent of India's GDP. Close to 550 sugar mills intermediaries currently operate in 16 states across the country, although 80% of national sugar output is concentrated in the three major producing states of Maharashtra, Uttar Pradesh and Karnataka. Figure 1 shows the intensity of cane cultivation and location of sugar mills across India in 2015-16.

Figure 1: Sugar Production in India



Panel A: Cane Cultivation, 2015-16 **Panel B:** Location of Sugar Mills, 2015-16

Notes: Panel A shows block-level share of cropped area under cane cultivation in 2015-16. Panel B shows the location of operational sugar mills in India in the 2015-16 season. *Source:* Indian Agriculture Census 2015-16, and plant-level records from the Indian Directorate of Sugar.

2.1 The Indian Sugar Value Chain

The Indian sugar value chain is visualized in Figure 2. At the top of the chain, 8 million smallholder farmers cultivate cane on plots averaging 0.8 hectares [12]. These farmers

[11] For details, see https://www.pib.gov.in/FactsheetDetails.aspx?Id=150971

[12] This figure comes from the Indian Agriculture Census 2015-16.

sell their cane to approximately 550 sugar mills, with each mill employing close to 1,000 workers and sourcing cane from nearly 15,000 farmers on average. In this setting, sugar mills therefore become the primary processing intermediaries that cane farmers transact with[13]. Within mills, cane is crushed to extract juice, and processed into raw or refined sugar. The sugar is then sold to wholesalers and retailers who supply the consumer market.

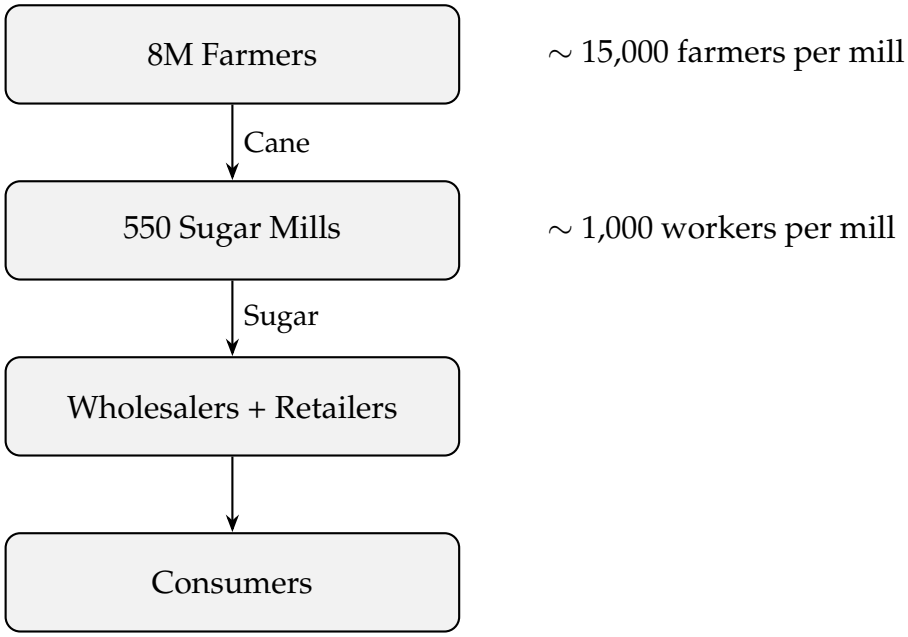


Figure 2: The Indian Sugar Value Chain

Mill ownership takes two primary forms: cooperatives and private enterprises. Cooperatives, which are particularly prevalent in Maharashtra and Karnataka, are nominally owned by their farmer-members but managed by elected boards. A natural concern is that cooperatives may face different optimization objectives than private mills, for instance, maximizing member welfare rather than profits (Hansmann, 2000). However, farmer-level data from Maharashtra in Figure A.1 reveals that less than a quarter of cane suppliers to a typical cooperative are registered member-farmers; the remaining three-fourths are non-member suppliers with no ownership stake. Since cooperatives must offer competitive cane prices to attract this large non-member majority who face the same outside options as farmers supplying to private mills, the pricing behaviour of cooperatives in the cane market is likely to resemble that of profit-maximizing private mills. As

[13]Over 80% of all cane produced is utilized for sugar production, with the rest sold to small jaggery producing units or used for self-consumption. I introduce these as an outside option in the inner nest of my nested discrete choice farmer supply model in Section 6.

such, the model that I develop in Section 6 treats ownership status as an observable characteristic entering farmer mill-choice but does not endogenize the governance structure.

2.2 Mill Control over the Harvest

A defining feature of the Indian sugar industry is that it is the mill, and not the individual farmer, that designs and coordinates the harvesting schedule during the crushing season, which typically runs for five months from November to April. Mills determine which fields are cut on which days to coordinate the movement of cane from field to factory. This arrangement is common across agro-processing crops whose quality degrades rapidly after cutting – tea, cotton, and cane alike. In such settings, the processor centralizes harvest logistics which helps ensure that the raw material is processed soon after harvest. The stakes are particularly high for cane because its sugar content is time-sensitive at both ends: cane accumulates sucrose as it matures, and begins to lose it through inversion once cut. The two levers that the mill controls – the *maturity* of the cane it schedules for harvest and the *time gap between cutting and crushing* – therefore directly shape how much sugar is ultimately extracted.

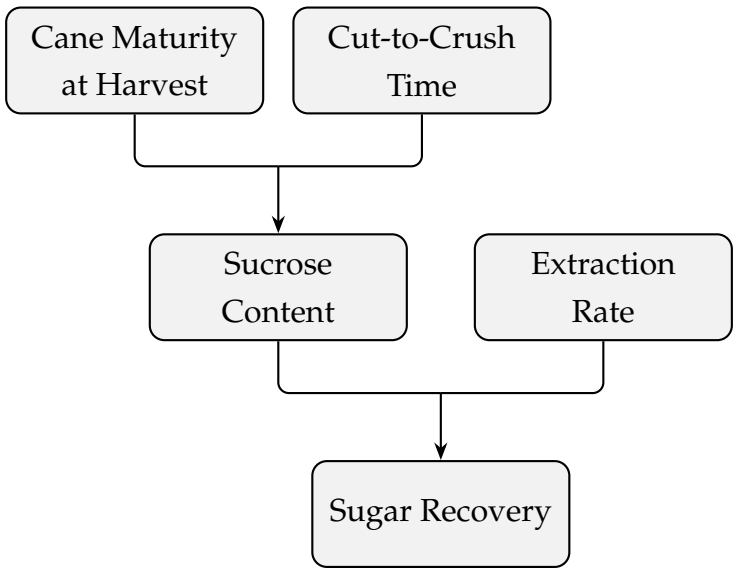
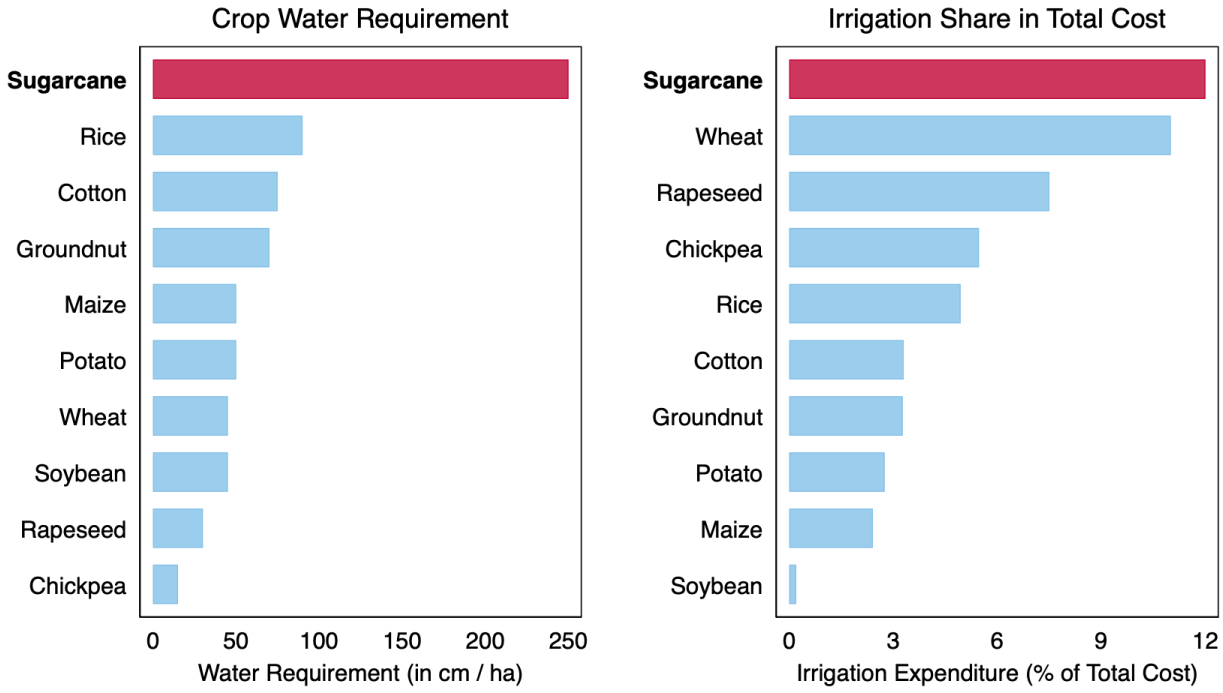


Figure 3: Determinants of Sugar Recovery

These harvest decisions are summarized by *sugar recovery*, the share of sugar extracted per unit of cane crushed, which serves as an observable measure of mill productivity throughout the paper. Figure 3 illustrates how the operational levers in the mill's control map into recovery: scheduling cane harvest at higher maturity and shortening cut-to-crush intervals raises its sucrose content. Combining these process management practices

with sucrose extraction pins down sugar recovery. Because recovery is reported mill-by-mill and season-by-season in administrative records, it provides a direct, high-frequency window into how effectively each mill manages its harvest – a productivity margin that, as I show later, responds to the competitive and climatic pressures at the heart of this paper.

2.3 Water Intensity and Sensitivity to Droughts



Panel A: Crop Water Requirement

Panel B: Irrigation Share in Total Cost

Figure 4: Cane's water intensity makes sugar mills vulnerable to droughts

Notes: Panel A shows crop water requirement over the growing season for 10 major crops, while Panel B shows their share of total cultivation cost spent on irrigation. *Source:* Indian National Bank for Agriculture and Rural Development (NABARD), and the Cost of Cultivation Surveys.

The Indian sugar industry's vulnerability to droughts stems from the extraordinary water intensity of cane cultivation. Cane requires approximately 250 centimetres of water per hectare over its 12 month growing cycle (Panel A of Figure 4), making it the most water-demanding crop in Indian agriculture. For comparison, sugarcane's water requirement is nearly thrice that of rice, the next most water-intensive major crop. Cane also stands out in terms of expenditure incurred on irrigation, with 12% of its total cost of cultivation

spent on irrigation through groundwater or canals (Panel B of Figure 4). This is despite the fact that a sizeable part of its growing cycle overlaps with the monsoon season, as opposed to wheat which is grown during the dry season. Crucially, the water intensity extends beyond cultivation into the processing stage. Sugar mills, on average, require 20 cubic metres of water to produce one metric tonne of sugar (Figure A.2). Both the upstream cultivation and downstream processing stages of the supply chain are thus acutely dependent on water, making the entire industry highly sensitive to droughts. This sensitivity is not merely theoretical – it is a recurring feature of India's economic landscape that regularly makes national headlines as can be seen in Figure A.3.

3 Data

I assemble four datasets that jointly capture mill-level operational outcomes, cane input prices, farmer supply behaviour, and drought shocks at resolutions fine enough to identify causal effects within local sugar markets. Three novel data contributions underpin the analysis. First, I digitise archival plant-level records from the Indian Directorate of Sugar - the closest available census of the Indian sugar manufacturing sector - covering the universe of all sugar mills from 1990 to 2025 and yielding a panel of unprecedented length and coverage. Second, I obtain first-time research access to the complete universe of farmer-mill cane transaction records maintained by the Maharashtra Sugar Commissionerate, providing georeferenced data on approximately 3 million unique cane farmers. Third, I develop a novel matching algorithm to link the Directorate's plant-level records to the Annual Survey of Industries (ASI), enabling a full mill-level price and financial analysis for the first time in the Indian sugar industry literature. This section describes each dataset in turn.

3.1 Plant-Level Data

3.1.1 Archival Mill-Level Records

The core of the database is a mill-level panel constructed from the Yearbook and Directory of Indian Sugar Factories, published annually by the Indian Directorate of Sugar. The Yearbook constitutes the closest available census of the Indian sugar manufacturing sector, covering all registered and operational mills in every crushing season. I obtain access to and digitise these archival records spanning 1990 to 2025 through an institutional relationship developed with the Directorate. I supplement these with plant-level records from India's two largest sugar producing states - Maharashtra and Uttar Pradesh - that

together account for over 70% of the country's sugar output. Records obtained from the state-level sugar departments provide an additional cross-check on coverage. The final sample consists of 886 firms in 750 plant locations across 20 states.

For each mill-year, the panel records administrative and manufacturing data for each step of the production process. Key variables of interest include total cane procured, sugar produced, crushing duration, installed capacity, capacity utilisation, cane maturity at harvest, sugar content in cane, overall sugar recovery rate, extraction efficiency, and water use. The sugar recovery rate - defined as sugar produced per unit of cane crushed, and expressed as a percentage - is the observable measure of mill-level productivity. Crucially, recovery can be decomposed into two independent components: Pol% (which measures sugar content in cane - a field-level variable reflecting variety choice and harvest timing), and Mill Extraction and Boiling House Recovery (which measure extraction efficiency - a factory-level technology variable). This decomposition allows the analysis to separately identify whether recovery gains arise from field-level managerial improvements or factory-level technological investments - the key distinction for the paper's mechanism. Operational status is coded as working, suspended, or permanently exited, with permanent exit assigned when a mill is non-working for more than three consecutive years with no subsequent resumption, or when corporate records confirm acquisition or liquidation.

3.1.2 Cane Variety Technology

The panel also records the varieties of cane procured by each mill in each season, providing a direct measure of varietal composition over time. I link these variety names to agronomic attributes – potential sucrose content, yield, maturity class, and agro-climatic zone using variety performance data from the All India Coordinated Research Project (AICRP) on sugarcane and Vasantdada Sugar Institute's variety evaluation database. Drought and stress tolerance classifications are obtained from the ICAR-Indian Sugarcane Research Institute. This enables construction of an annual mill-level cane variety index.

3.1.3 Online Mill Survey

To supplement the administrative records with information on management practices and organizational structure not captured in the panel, I conduct an online survey of 220 sugar mills in Maharashtra, circulated through the Maharashtra Sugar Commissionerate. The survey covers drought risk and adaptation, harvest scheduling practices, extension staff deployment, seed distribution programmes, and recent capital investments - the management inputs that the model identifies as the channel for within-firm productivity

upgrading.

3.2 Financial Data

3.2.1 Annual Survey of Industries

Cane input prices and financial structure at the mill level are drawn from the Annual Survey of Industries (ASI), conducted annually by India's Ministry of Statistics and Programme Implementation. The ASI covers all factories registered under the Factories Act, 1948, and provides data on revenues, costs, wages, capital, material inputs, and output - making it the principal source of industrial statistics in India and the only source providing a standardized, all-India financial panel for manufacturing plants.

A central challenge is that the ASI does not identify factories by name; it uses an internal factory code not publicly linked to plant identities, and these codes change with ownership transfers or re-registration. I overcome this challenge through a matching algorithm that exploits variables common to both the archival plant records and ASI. This yields a matched panel for the near-universe of sugar mills from 1993-2023, enabling a full cane price and financial analysis at the individual mill level for the first time in the Indian sugar industry literature.

3.2.2 Maharashtra Financial Data

For Maharashtra, three additional sources provide financial data. First, the Sugar Commissionerate publishes annual mill-level price reports, recording the actual cane price paid to farmers by season. Second, the Vasantdada Sugar Institute maintains detailed cost-of-production records for sugar mills in Maharashtra from 1998–2023, serving as an independent validation layer for ASI-derived cost variables and as a primary source for mills outside the ASI match. Third, I draw on mill licence application records maintained by the Commissionerate covering 1990-2025. These records document the year of application, the year of first crushing, and for mills that entered, directly observed entry costs decomposed into land acquisition, machinery, and building costs. This is, to my knowledge, the only source providing directly observed sunk entry costs for sugar mills in India, enabling identification of the entry cost parameter from data rather than from equilibrium restrictions alone.

3.3 Farmer-Mill Transactions Data

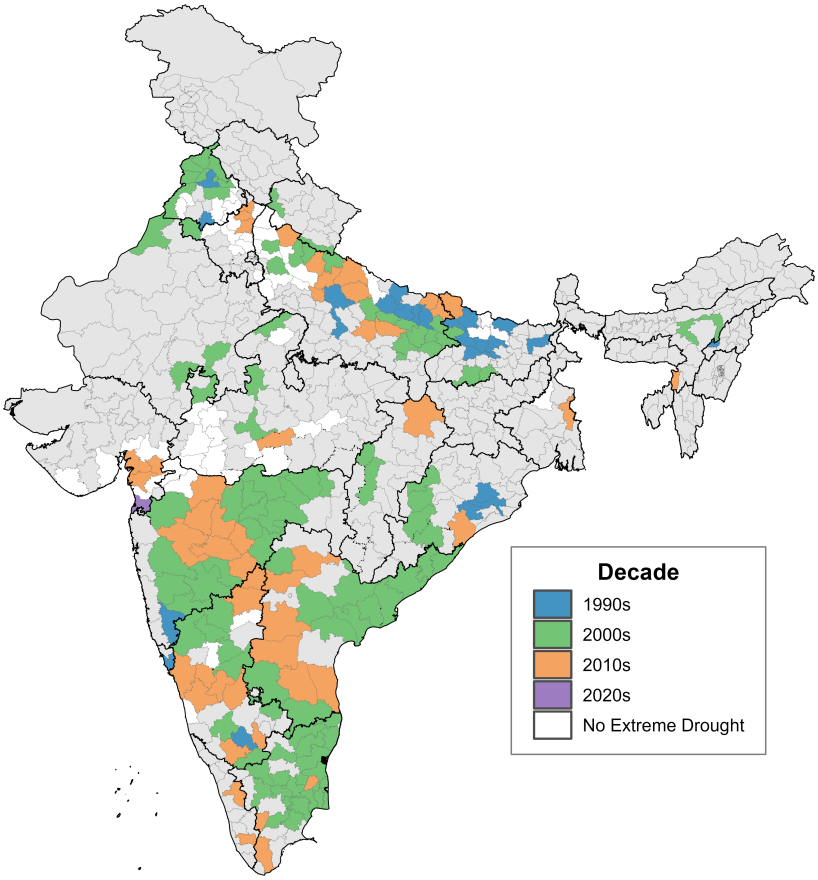
The Maharashtra Sugar Commissionerate maintains, under the state's mandatory cane registration system, a complete record of every farmer-mill cane transaction in Maharashtra. For each crushing season from 2020 to 2025, the records document, for every farmer supplying cane to every mill: the farmer's village, cane variety grown, area under cane, maturity class, and cooperative membership status. I georeference these records by matching Commissionerate village names to census village identifiers through a multi-stage fuzzy matching pipeline. The final output covers approximately 3 million unique farmer observations across 220+ mills and 16,000 villages.

3.4 Weather Data

The treatment variable throughout the main analysis is an indicator for an extreme drought at the district level, constructed from the Drought Atlas of India (Chuphal et al., 2024) - a high-resolution dataset providing monthly Standardised Precipitation Evapotranspiration Index (SPEI) values across India from 1901 to 2021. The SPEI accounts for atmospheric water demand, temperature-driven evapotranspiration, alongside water supply, making it better suited to capturing crop-water stress than simpler precipitation measures such as SPI. The SPEI values are constructed using the Hargreaves method, which requires only minimum and maximum temperature and extraterrestrial radiation as inputs, and is considered highly reliable for Indian climatic conditions.

I use a 12-month accumulated SPEI measured from January to December, chosen to match the sugarcane cultivation cycle: cane is typically planted in January and harvested in December, so a January-to-December window captures the full annual water balance relevant to the crop. An extreme drought is defined as a district-year in which SPEI falls at or below -2, corresponding approximately to a 1-in-50 year event.

Figure 5: Drought Incidence across Sugar Producing Districts



Construction proceeds in three steps. First, I aggregate the Drought Atlas's native-resolution data to the block level within each district. Second, if any block within a district records SPEI ≤ -2, that district is classified as drought-affected in that year. This is justified by the fact that mills source cane from catchment areas of approximately 50 kilometres, so a block-level extreme drought will affect all mills in the district. Third, to avoid complications from multiple drought incidences within the same district over time, I retain only the single most severe drought for each district. Severity is defined as the year in which the largest share of the district's area falls below SPEI ≤ -2 - that is, the drought with the broadest spatial coverage. This ensures each treated district has one cleanly defined treatment year, simplifying the event study design.

Over the sample period, the data records 174 extreme drought events affecting 226 distinct districts across 20 cane-producing states, with staggered timing across 25 distinct years that provides the identifying variation. Figure 5 maps this spatial and temporal incidence of extreme droughts across Indian districts, colour-coded by the decade in which

each district experienced its most severe event. The map reveals rich variation across both space and time: drought-affected districts are distributed across all major sugar-producing regions, with no single state or region bearing the shock exclusively.

3.5 Supplementary Data

I complement the core datasets with several agricultural, irrigation, and geographic sources used for controls, the farmer supply model, and robustness checks. The quinquennial Agriculture Census provides block- and district-level crop area and irrigated area. The Minor Irrigation Census rounds provides data on wells, tube well depths, and irrigated command area, used as controls for groundwater dependence. Farmer-level Cost of Cultivation Surveys (2000-2020) provide input costs, revenue and yields for major crops including sugarcane. The ICRISAT District-Level Database (to 2017), supplemented by Area, Production and Yield (APY) reports from the Directorate of Economics and Statistics for subsequent years, provides district-level crop yields, output, acreage, and prices for all major crops entering the Fréchet outer nest. FAO-GAEZ potential crop yields under different agro-climatic conditions are used to construct land suitability parameters in the supply model. Finally, Central Ground Water Board's (CGWB) observation-well-level groundwater depth measurements are used in the event study and as a control for irrigation dependence.

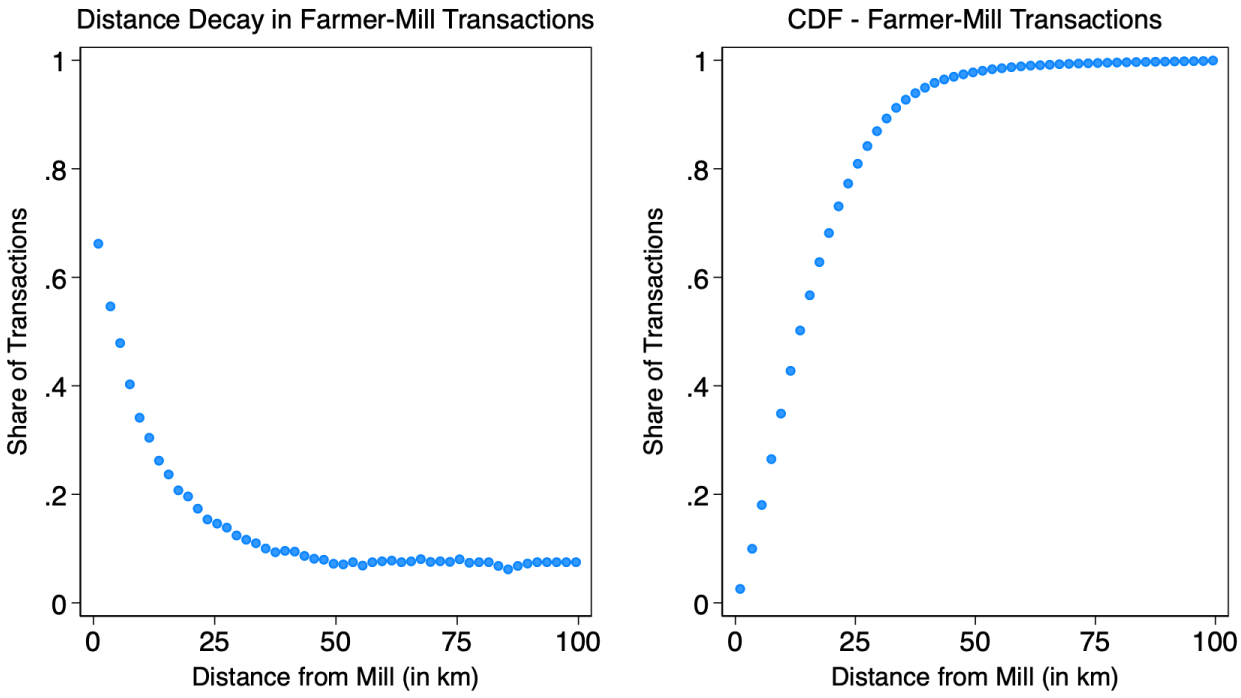
4 Stylized Facts

Before turning to the event study results, I present three stylized facts that establish the institutional conditions under which the paper's mechanism operates.

Fact 1: Trade in cane is highly localized, generating spatially segmented markets.

Cane markets are spatially segmented to a degree that is unusual even among agricultural commodities. Using georeferenced farmer-mill transaction data, I document extreme distance decay in farmer-mill relationships. Figure 6, panel A plots the share of farmers selling to a given mill as a function of the distance between them. Among farmers located within 5 kilometres of their closest mill, 70 percent sell their cane to that mill, and this share drops to less than 10 percent by 50 kilometres. Figure 6, panel B presents the cumulative distribution function: roughly 80 percent of all farmer-mill transactions take place within 25 kilometres of a mill and virtually all trade in cane exhausts within 50 kilometres of a mill.

Figure 6: Trade in cane exhausts within 50 kms of a mill



Panel A: Distance Decay

Panel B: CDF - Distance Decay

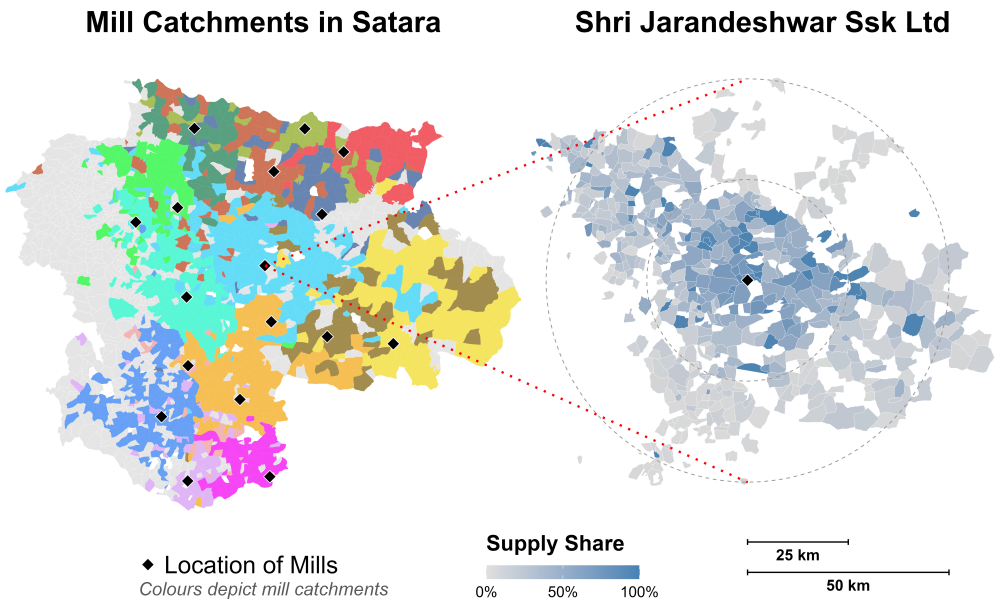
Notes: Panel A shows the share of farmers selling to a given mill as a function of the distance between them. Panel B plots the CDF. *Source:* Farmer-Mill Transactions from the Maharashtra Sugar Commissionerate.

This extreme spatial concentration arises from the rapid post-harvest deterioration of sugar content in cane. Once cut, cane loses recoverable sugars rapidly due to microbial activity and chemical inversion of sucrose into non-recoverable reducing sugars – glucose and fructose. This perishability creates strong incentives for mills to source cane from nearby farms and crush cane within 12-18 hours of harvesting. Such a highly localized nature of trade in cane gives rise to spatially segmented markets as illustrated in Figure 7. Panel A maps the district of Satara in Maharashtra, with each village colour-coded by the mill that receives the largest share of its cane and reveals sharp spatial partitioning. Panel B zooms into the catchment area of a single mill, showing that the mill's procurement share across surrounding villages decays steeply with distance – consistent with the distance gradient documented in Figure 6.

This fact has three implications for the paper. First, it defines the relevant market: cane markets are local, and market concentration measures defined at the district level capture meaningful variation in competitive conditions. I therefore adopt the district as the unit

of a local market throughout the analysis. This choice is validated by farmer-mill transaction data from Maharashtra, which shows that cross-district trade in cane accounts for only 10 percent of all transactions (Figure A.4) – the vast majority of cane procurement occurs within district boundaries. Second, it ensures that when a mill exits, its farmers are reallocated to nearby surviving mills. Third, it creates oligopsony power: with only a fixed set of mills within farmers' economical transport distance, each mill faces a local supply curve with limited elasticity, giving it price-setting power over cane procurement.

Figure 7: Cane markets are spatially segmented



Notes: The plot on the left shows the distribution of mill catchments in the district of Satara, Maharashtra. The plot on the right zooms into the catchment for one mill, showing the mill's procurement share across supplying villages in a 50 km radius. *Source:* Farmer-Mill Transactions from the Maharashtra Sugar Commissionerate.

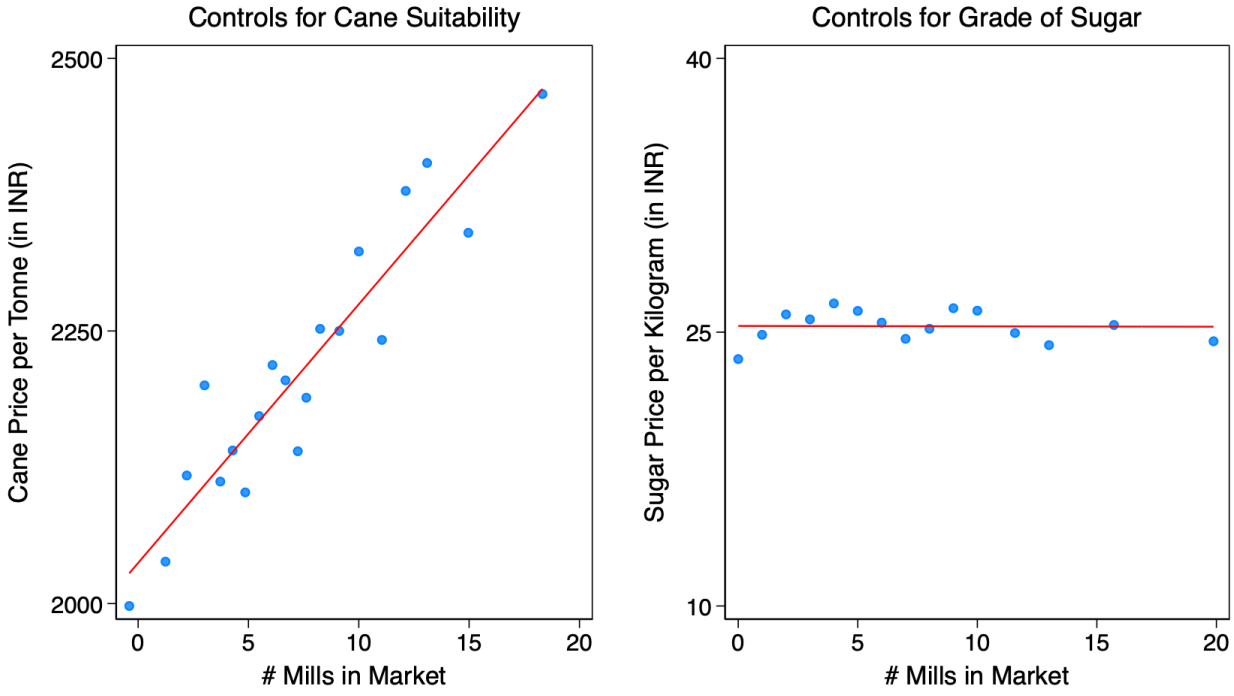
Fact 2: Mills set cane prices, but are price-takers in sugar markets.

Cane markets are spatially segmented because cane is bulky, perishable, and costly to transport. This feature creates local oligopsonies: a given farmer's choice set is limited to the few mills within economical transport distance. To protect farmers, cane prices are subject to regulatory price floors in the form of a centrally determined Fair and Remunerative Price (FRP) and, in several states, the higher State Advised Price (SAP).[14]

[14]Four states - Uttar Pradesh, Punjab, Haryana and Uttarakhand set their own State Advised Price (SAP), which specifies two to three distinct prices according to the cane variety a farmer grows; the remaining states follow the centrally determined Fair and Remunerative Price (FRP), which combines a baseline price with a premium that increases in the mill's sugar recovery.

However, these floors are not binding in practice: more than 90% of all observations in the data see farm-gate cane prices above the price floor. Panel A of Figure 8 demonstrates the resulting oligopsony power: cane prices paid to farmers are higher in markets with a greater number of mills, after controlling for cane quality and suitability. Moving from near-monopsony to a competitive market of 15 mills raises cane prices by approximately 25 percent, suggesting price-setting power by mills in cane markets.

Figure 8: Mills have price-setting power in cane markets



Panel A: Cane Prices

Panel B: Sugar Prices

Notes: Panel A shows a binscatter plot for farm-gate cane prices against local market structure after adjusting for cane quality and suitability. Panel B shows a binscatter plot for ex-mill sugar prices against local market structure after adjusting for the grade of sugar produced. Data is for year 2010. *Source:* Plant-level Records from the Indian Directorate of Sugar and the Annual Survey of Industries.

On the output side however, the picture is starkly different. Sugar prices are effectively determined by government regulation through a system of monthly release quotas that control how much sugar each mill can sell in a given month. An example of such a monthly sugar release allocation can be seen in Figure A.5 for sugar mills in Punjab. Since each mill's saleable quantity is administratively fixed, no individual mill can influence the market price by expanding or withholding supply. Panel B of Figure 8 confirms this: ex-mill sugar prices in the year 2010 show no systematic relationship with local market

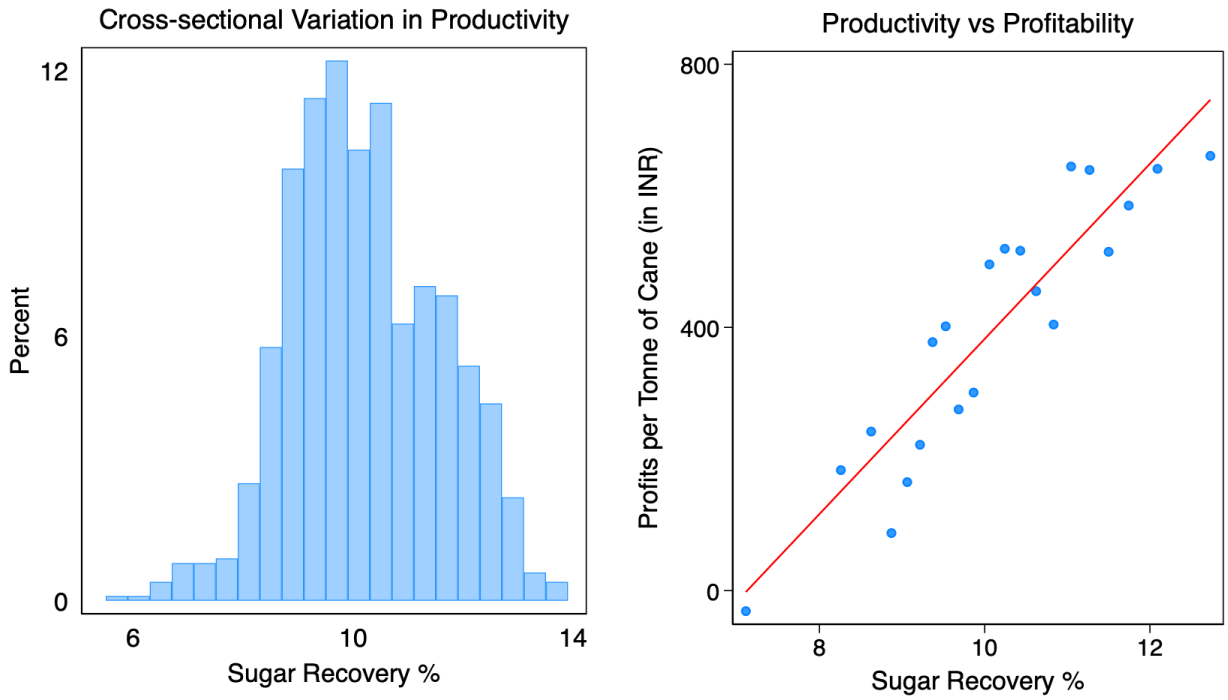
structure after controlling for the grade of sugar produced. Sugar prices are seen hovering around Rs. 25-26 per kilogram regardless of how competitive a mill's local market is, as measured by the number of mills located in the district. Market power, absent on the output side but pronounced on the input side, is a defining feature of the industry and motivates the oligopsony model I develop in Section 6.

Fact 3: Mills vary widely in productivity, and the least productive operate on the thinnest margins.

I use sugar recovery as an observable measure of mill productivity and find that mills differ enormously in how efficiently they convert cane into sugar. Figure 9, panel A plots the distribution of sugar recovery across mills and reveals substantial cross-sectional dispersion. Recovery ranges from roughly 6 to 14 percent, implying that the least productive mills extract less than half as much sugar per tonne of cane as the most productive. Because sugar revenue is linear in recovery, differences of this magnitude translate into large differences in the value a mill generates from the same quantity of cane. This productivity dispersion maps directly into profitability. Panel B plots profit per tonne of cane against sugar recovery, and shows a strong positive gradient. The least productive mills operate on the thinnest margins, while high-recovery mills earn substantially more per tonne crushed. The relationship is intuitive – a low-recovery mill extracts less sugar from each tonne of cane it purchases, yet pays a cane price anchored by the regulated floor and by competition for cane as described in Fact 2, so its margin is compressed from both sides. Low recovery is therefore not merely a technical inefficiency, it is a direct determinant of financial fragility.

This fragility is what makes productivity dispersion consequential for how the industry responds to drought shocks. Droughts sharply contract cane availability for up to two years after the drought itself. When cane becomes scarce, every mill faces lower throughput, but thin-margin mills have the least buffer to absorb a prolonged period of underutilized crushing capacity. The least productive mills are thus the most exposed to drought. Already operating close to the break-even point, they are the first to become unviable when a shock decimates local cane supply.

Figure 9: Mills differ in productivity, with laggards facing thinner margins



Panel A: Sugar Recovery

Panel B: Profits

Notes: Panel A shows a cross-sectional histogram on sugar recovery. Panel B shows a binscatter plot for sugar recovery against profits per tonne of cane. *Source:* Plant-level Records from the Indian Directorate of Sugar and the Annual Survey of Industries.

Taken together, the three facts establish the conditions for the paper's mechanism. **Fact 3** identifies which mills are at the margin of survival, and hence which mills a drought is likely to force into exit – a selection that is central to the restructuring mechanism I study. **Fact 1** implies that drought-induced reallocation is not random across space: it removes the least productive mills and reallocates their cane to local survivors, reshaping local market structure and, through the oligopsony power documented in **Fact 2**, the prices farmers ultimately receive. This selection-and-reallocation mechanism is the focus of the model developed in Section 6. The next section presents the event study evidence that these ingredients produce.

5 Empirics

5.1 Empirical Strategy

My identification strategy exploits spatial and temporal variation in the incidence of extreme drought events across India's sugar-producing districts. Extreme droughts are defined as district-year observations where the Standardised Precipitation Evapotranspiration Index (SPEI) falls below -2, corresponding to a 1-in-50 year event. Because sugar production is geographically dispersed across India, different districts experience extreme droughts at different times, providing quasi-experimental variation in the external shock. To analyse the effects of these drought shocks, I implement an event-study design at two levels:

At the **market level**, I use the following specification:

$$y_{dt} = \sum_{k=\underline{T}}^{\bar{T}} \beta_k D_{dt}^k + \delta_d + \lambda_{zt} + \varepsilon_{dt} \tag{1}$$

where y_{dt} is the outcome for district d in year t, δ_d are district fixed effects absorbing time-invariant market characteristics, λ_{zt} are cane climatic zone-year fixed effects absorbing common annual shocks within cane zones, and D_{dt}^k are event-time indicators defined relative to the year in which district d experiences its worst extreme drought.

At the **plant level**, I use the following specification:

$$y_{it} = \sum_{k=\underline{T}}^{\bar{T}} \theta_k D_{it}^k + \delta_i + \lambda_{zt} + u_{it} \tag{2}$$

where y_{it} is the outcome for mill i in year t, δ_i are mill fixed effects, λ_t are cane climatic zone-year fixed effects. Mill-level treatment assignment is based on the drought status of the district in which the mill is located. The coefficient for the event year ($k = 0$) is normalised to zero. Both specifications contain lags and leads from -8 to +12 years around the event respectively, but I restrict the figures to show treatment effects from -5 to +10.

Given the staggered implementation of droughts across districts, I estimate all dynamic treatment effects using the Sun and Abraham (2021) estimator, which addresses concerns about bias from heterogeneous treatment effects in staggered difference-in-differences

settings. For robustness, I also estimate using standard two-way fixed effects and show that the results are very similar.

The key identifying assumption is that, in the absence of droughts, outcomes for mills (or districts) in affected areas would have followed similar trends as those in unaffected areas, conditional on mill (or district) and zone-year fixed effects. The features supporting identification are: (i) the incidence of extreme drought is plausibly exogenous to local economic conditions, being driven by large-scale climatic phenomena as described in Section 3; (ii) the spatial variation across districts experiencing drought in different years provides multiple treatment cohorts, with approximately a quarter of the districts never experiencing an extreme drought during the sample period, serving as a permanent control group; and (iii) the pre-treatment event-study coefficients are statistically indistinguishable from zero, supporting the parallel trends assumption.

5.2 Markets

5.2.1 Droughts contract cane supply for up to 2 years

5.2.2 Droughts cause the least productive mills to exit and their farmers to reallocate to nearby survivors

5.3 Sugar Mills

5.4 Cane Prices

5.5 Cane Income Decomposition

5.6 Summary of Empirical Results

The reduced-form evidence establishes five results that, taken together, trace the causal chain from drought-induced restructuring to within-firm upgrading:

These five results motivate and discipline the structural model developed in the next section. The model must rationalise why marginal survivors, who receive the cane volume windfall, exhibit the strongest upgrading response. It must also explain why these productivity gains are not passed through to farmers, and quantify the welfare trade-off between aggregate productivity improvements and distributional costs.

6 Model

6.1 Intermediary Cane Demand

6.1.1 Technology

Sugar mill i in year t produces sugar Q_{it} using cane M_{it}, labour L_{it}, which are both expressed in quantities, and fixed assets K_{it}. I assume that cane cannot be substituted with either labour or capital, so production is Leontief:

$$Q_{it} = \min \left\{ \beta_{it}^{M} M_{it}, \;\; \Omega_{it} H(L_{it}, K_{it}) \right\}. \tag{3}$$

The first arm corresponds to the amount of sugar recovered directly from cane: a tonne of cane yields β_{it}^{M} tonnes of sugar, where β_{it}^{M} is *sugar recovery*, the observable measure of mill productivity. The second arm is a labour-capital aggregate $H(\cdot)$, twice differentiable in both inputs with common parameterisation β, scaled by Hicks-neutral productivity Ω_{it}. Processing capacity is bounded: cumulative crushing cannot exceed $\bar{M}_{it}$, the mill's maximum seasonal throughput, which I construct as the mill's installed crushing capacity (tonnes crushed per day) multiplied by the length of the crushing season in its state (days), inferred from manufacturing records:

$$M_{it} \;\leq\; \bar{M}_{it} \;\equiv\; \underbrace{\kappa_{it}}_{\text{Installed Capacity (TCD)}} \;\times\; \underbrace{D_{s(i)t}}_{\substack{\text{Season Length} \\ (days)}} \tag{4}$$

Departing from the common assumption that the conversion rate is homogeneous across firms (Rubens, 2023; Avignon and Guigue, 2025) or exogenous (De Loecker and Scott, 2016), I treat recovery as a plant-level *choice* which can be improved through cane-quality upgrading, motivated by the empirical results shown in Section 5. Upgrading is a discrete management decision $u_{it} \in \{0, 1\}$: a mill that adopts the cane-quality upgrading programme raises recovery by a factor $\gamma > 1$, at a fixed cost F_{up},

$$\beta_{it}^{M}(u_{it}) = \left(1 + (\gamma - 1)\, u_{it}\right) \beta_{it}^{M} \;=\; \begin{cases} \beta_{it}^{M} & \text{operate as normal } (u_{it} = 0), \\ \gamma\, \beta_{it}^{M} & \text{upgrade } (u_{it} = 1). \end{cases} \tag{5}$$

Treating the conversion rate as heterogeneous and firm-chosen is the key institutional feature of agricultural processing, not only in sugar but in coffee, cotton, tea, and other

goods, something that a homogeneous-recovery model omits. Equation 3 nests production functions in which all inputs are substitutable; the recovery parameter would then be zero, and cane would be an additional substitutable input in $H(\cdot)$.

I assume that cane and labour are variable inputs: their amount is chosen by mills in the same time period as when they are used. I also assume that cane and labour are static inputs, which means that they only affect current profits. These assumptions rule out adjustment costs over sugar and inventories. Capital is, in contrast, dynamic and fixed; the capital stock at time t can only be changed through investment at time t - 1.

6.1.2 The Mill's Problem

Each year, mills choose labour L_{it}, cane M_{it}, and the discrete upgrading decision $u_{it} \in \{0, 1\}$ to maximise current variable profits. Because labour and cane are complements, choosing labour pins down cane and hence output. Through their cane quantity choices, mills also set cane prices subject to a floor. Hence, cane farmers are assumed to be price-takers, which is intuitive given that there are many more farmers than sugar mills (∼15,000 farmers per mill). Mills are price-takers in sugar and labour (P_t^s, W_{it}^L) and oligopsonists in cane.

$$\Pi_{it} = \max_{L_{it},\, M_{it}} \max \Big\{ \underbrace{P_t^s \beta_{it}^M M_{it} - W_{it}^M M_{it} - W_{it}^L L_{it} - F_{\text{op}}}_{\text{operate as normal}},\ \underbrace{P_t^s \gamma \beta_{it}^M M_{it} - W_{it}^M M_{it} - W_{it}^L L_{it} - F_{\text{op}} - F_{\text{up}}}_{\text{upgrade } (\gamma>1,\ \text{pay } F_{\text{up}})} \Big\}, \tag{6}$$

where sugar revenue uses $Q_{it} = \beta_{it}^M M_{it}$ from (3), and F_{op} is a fixed operating cost. The problem is subject to the capacity constraint and the statutory price floor, which has base price ϕ_0 at reference recovery $\bar{\beta}^M$ and a recovery premium $\phi_1 > 0$:

$$M_{it} \;\le\; \bar{M}_i, \qquad W_{it}^M \;\ge\; \phi_0 + \phi_1 \big(\tilde{\beta}_{it}^M - \bar{\beta}^M\big), \tag{7}$$

where $\tilde{\beta}_{it}^M \in \{\beta_{it}^M, \gamma\beta_{it}^M\}$ is the mill's realised recovery. The floor is the Fair and Remunerative Price (or, in SAP states, the higher State Advised Price); its recovery-linked premium means an upgrading mill faces a higher mandated price. A mill operates only if its variable profit is non-negative; otherwise it exits ($\pi(\beta_{it}^M) < 0 \Rightarrow$ exit), withdrawing its cane demand and, as shown below, widening the markdowns of surviving rivals.

6.1.3 Input Prices and Markdowns

Solving the cane first-order condition delivers the cane-price markdown:

$$W_{it}^M = \frac{\varepsilon_{it}^s}{\varepsilon_{it}^s + 1}\Big(\text{NetMRP}_{it}^M - \lambda_{it}\Big), \qquad \text{NetMRP}_{it}^M \equiv \tilde{\beta}_{it}^M\Big(P_t^s - \frac{W_{it}^L}{\Omega_{it} H_L}\Big), \tag{8}$$

where NetMRP_{it}^M is the marginal revenue product of cane net of the complementary labour cost of processing it, $\lambda_{it} \geq 0$ is the shadow value of the capacity constraint with $\lambda_{it}(\bar{M}_i - M_{it}) = 0$, and $H_L \equiv \partial H / \partial L$. The factor $\varepsilon_{it}^s/(\varepsilon_{it}^s + 1) < 1$ is the oligopsony markdown: a less elastic residual cane supply (lower ε_{it}^s) widens the markdown – for instance when drought-induced exit concentrates procurement on a surviving mill. The capacity multiplier depresses the price further: a mill at its crushing limit values marginal cane only up to its processing value, so $\lambda_{it} > 0$ lowers W_{it}^M. When the floor binds the mill instead pays $W_{it}^M = \phi_0 + \phi_1(\tilde{\beta}_{it}^M - \bar{\beta}^M)$.

6.1.4 The Upgrading Decision

Upgrading is discrete: a mill upgrades iff the upgrade payoff in (6) weakly exceeds the operate-normally payoff. When the floor is slack the mill pays the markdown $W_{it}^M = \psi_{it} P_t^s \tilde{\beta}_{it}^M$ with $\psi_{it} \equiv \varepsilon_{it}^s/(\varepsilon_{it}^s + 1)$, keeping only the $(1 - \psi_{it})$ share of the value it creates; the labour and operating terms cancel and the decision reduces to

$$(1 - \psi_{it})\, P_t^s\, (\gamma - 1)\, \beta_{it}^M\, M_{it} \;\geq\; F_{\text{up}} \quad \Longleftrightarrow \quad M_{it} \;\geq\; M_{it}^* \equiv \frac{F_{\text{up}}}{(1 - \psi_{it})\, P_t^s\, (\gamma - 1)\, \beta_{it}^M} \tag{9}$$

Two implications follow.

(i) **Scale-Upgrading Complementarity.** The upgrade benefit is increasing in cane volume, so only mills above the threshold M_{it}^* upgrade. Reallocation of cane towards survivors raises their scale and pushes them past the threshold, so absorbing mills upgrade and aggregate recovery rises with concentration.

(ii) **Buyer Power.** The benefit also scales with the captured share $(1 - \psi_{it})$, which is larger when residual supply is less elastic. Greater buyer power lowers M_{it}^*, so the same restructuring that raises a survivor's scale also raises the share of the upgrading gain it retains.

Market restructuring thus raises both the incentive to upgrade (scale M) and the power to capture the gains (the markdown share $1 - \psi$), pushing survivors past M^*.

6.1.5 Cane Input Demand

Inverting the labour condition under a slack price floor and slack capacity gives optimal labour, and substituting into (3) delivers the cane input-demand function which is truncated at $\bar{M}_i$ when capacity binds.

$$M_{it}^{*} = \frac{1}{\tilde{\beta}_{it}^{M}} \left(\Omega_{it} \left(\beta^{L}\right)^{\beta^{L}} K_{it}^{\beta^{K}} \left[\frac{P_t^s - \frac{W_{it}^{M}}{\tilde{\beta}_{it}^{M}} \left(\frac{\varepsilon_{it}^{s}+1}{\varepsilon_{it}^{s}} \right)}{W_{it}^{L}} \right]^{\beta^{L}} \right)^{\frac{1}{1-\beta^{L}}} \tag{10}$$

Compactly, $M_{it} = m\left(P_t^s, W_{it}^{M}, W_{it}^{L}, K_{it}, \beta^{L}, \beta^{K}, \tilde{\beta}_{it}^{M}, \Omega_{it}, \varepsilon_{it}^{s}\right)$, which is strictly monotone in productivity Ω_{it}. Except the production parameters $(\beta^{L}, \beta^{K}, \Omega_{it})$, all other parameters on the right-hand side $(P_t^s, W_{it}^{M}, W_{it}^{L}, L_{it}, K_{it}, \tilde{\beta}_{it}^{M})$ are observed, with ε_{it}^{s} coming from the upstream supply model. Monotonicity in Ω_{it} makes (10) the proxy that inverts for latent productivity in estimation using the control function approach of Ackerberg, Caves and Frazer (2015).

6.2 Upstream Cane Supply

Farmers face a nested discrete choice problem. In the inner nest, a farmer who has chosen to grow cane selects which mill to sell to. In the outer nest, the farmer decides whether to grow cane at all, or to allocate land to an alternative crop. The nesting structure ensures that mill exit affects farmers through two channels: it reduces the set of mills available (inner nest) and, through the inclusive value, reduces the attractiveness of cane relative to other crops (outer nest).

6.2.1 Environment

A market d (a district-year) contains a unit mass of land plots indexed by ω, with cultivable area $\bar{L}_d$. Each plot is operated by a farmer who allocates its use to the crop with the highest revenue per hectare. The choice set has two levels. At the outer level the plot is allocated to one of $K+1$ crops $k \in \{0, 1, \ldots, K\}$, where $k = 0$ denotes sugarcane and $k = K$ a reference crop. At the inner level, conditional on cane, the plot's output is sold to one of the mills i from the local choice set $I_d \cup \{0\}$, where $\{0\}$ denotes an outside option (selling to a jaggery producer).

For a competing crop $k \geq 1$, revenue per hectare is $\mathcal{R}_k(\omega) = \tilde{p}_k \, \eta_k(\omega)$, where $\tilde{p}_k$ is the crop's effective output price and $\eta_k(\omega)$ is the plot's idiosyncratic productivity. For cane

sold to mill i, revenue is $\mathcal{R}_{0i}(\omega) = P_i\,\eta_0(\omega)\,\nu_i(\omega)$, where η_0 is cane productivity, ν_i a plot–mill match shock, and P_i the value of mill i per efficiency unit of cane. I parameterize the log of this value as linear in observables,

$$\ln P_i \;\equiv\; \tfrac{1}{\theta}\,\delta_i, \qquad \delta_i(d,t) = \alpha \ln W_i^M(d,t) + \mathbf{x}_{it}'\boldsymbol{\gamma} + \xi_i(d,t), \qquad \delta_0 \equiv 0, \tag{11}$$

where W_i^M is the cane price offered by mill i, and $\mathbf{x}_{it}$ captures observable mill characteristics. The coefficient $\alpha > 0$ governs the price response, and the term $\xi_i(d,t)$ captures unobserved mill-specific amenity.

Nesting assumption. Competing crop productivities are independent Fréchet, $\Pr[\eta_k \le \eta] = \exp(-T_k\eta^{-\theta})$, with shape $\theta > 1$ and scale $T_k > 0$, independent across crops and plots. The cane revenues $\{\mathcal{R}_{0i}(\omega)\}_i$ are instead positively dependent within the cane nest: they share the common cane component $\eta_0(\omega)$, while the mill match shocks $\nu_i(\omega)$ are i.i.d. across mills. The strength of this within-nest dependence is governed by a dissimilarity parameter $\rho_0 \in (0,1]$ ($\rho_0 = 1$ being independence). I take this correlation as a primitive, built into the joint distribution of cane revenues rather than derived by adding shocks, consistent with what a nested GEV generator provides (McFadden, 1977). In logs, $\ln \mathcal{R}_{0i}$ is a deterministic term, an i.i.d. Gumbel mill shock at scale σ, and a common cane-nest component – a Cardell(ρ_0) variate (Cardell, 1997) – that ties the mills together. Since the log of a Fréchet variate is Gumbel, revenue maximization in levels is equivalent to additive random-utility maximization in logs, so the Fréchet outer nest and the Gumbel inner nest are two levels of a single nested-GEV problem. I exploit this equivalence throughout: it delivers closed-form shares and, as shown below, a testable cross-parameter restriction.

6.2.2 Inner Nest: Mill Choice Conditional on Growing Cane

Under the nesting assumption above, the probability that a farmer sells to mill i is:

$$s_{i|0}(d,t) = \frac{\exp(\delta_i/\sigma)}{\sum_{i' \in \mathcal{I}_d \cup \{0\}} \exp(\delta_{i'}/\sigma)} \tag{12}$$

where $\sigma > 0$ is the logit scale of the mill nest. The inclusive value of the cane nest summarises the expected payoff from the best available mill option:

$$IV_0(d,t) = \sigma \ln \left(\sum_{i \in \mathcal{I}_d \cup \{0\}} \exp(\delta_i/\sigma) \right) \tag{13}$$

This object has three properties that matter for the model. First, IV_0 is increasing in the number and quality of available mills: adding a mill strictly raises IV_0, and removing one (through exit) strictly lowers it. Second, IV_0 is increasing in any mill's cane price, since higher prices raise δ_i. Third, IV_0 links the inner nest to the outer nest: it enters the crop choice as the effective cane price, so that changes in the mill landscape - exit, entry, or price changes - propagate to farmers' crop allocation decisions.

6.2.3 Outer Nest: Crop Choice on Heterogeneous Land

In the outer nest, the farmer allocates land to a crop $k \in \{0 \text{ (cane)}, 1, \ldots, K\}$ with the highest revenue. Following Costinot, Donaldson and Smith (2016), each plot has a vector of crop-specific productivities drawn from independent Fréchet distributions with location parameters $T_k(d)$ (capturing land suitability for crop k in market d) and common shape parameter $\theta > 1$ (governing the dispersion of comparative advantage across plots). The share of land allocated to crop k is:

$$s_k(d,t) = \frac{T_k(d) \cdot \tilde{p}_k(d,t)^\theta}{\sum_{k'} T_{k'}(d) \cdot \tilde{p}_{k'}(d,t)^\theta} \tag{14}$$

where $\tilde{p}_k$ is the effective price of crop k. For cane, the effective price is derived from the inclusive value: $\tilde{p}_0(d,t) = \exp(IV_0(d,t)/\alpha)$, which aggregates over all mill options weighted by their attractiveness. For alternative crops, $\tilde{p}_k$ is the market price.

The Fréchet structure captures heterogeneity in land suitability: some plots are well suited to cane (e.g., irrigated, alluvial soils) while others are better for soybean or pulses. The shape parameter θ governs the aggregate acreage response to price changes: higher θ means more plots are on the margin between cane and alternative crops, so a given price change induces a larger shift in acreage.

The cane supply in market d to mill i is:

$$C_i(d,t) = \bar{L}_d \cdot y_0(d,t) \cdot s_0(d,t) \cdot s_{i|0}(d,t) \tag{15}$$

where $\bar{L}_d$ is the cultivable land area, y_0 is yield per hectare, s_0 is the cane acreage share (from the outer nest), and $s_{i|0}$ is the within-cane mill share (from the inner nest).

Consistency of the two nests imposes a restriction that is central to interpretation and identification. Requiring (12) and (14) to arise from a single valid nested-GEV generator

pins the dissimilarity to

$$\rho_0 = \frac{\theta\sigma}{\alpha} \in (0, 1], \qquad \text{equivalently} \qquad \frac{\alpha}{\sigma} \geq \theta. \tag{16}$$

Economically, mills must be closer substitutes for one another than cane is for competing crops – the within-cane substitution elasticity α/σ weakly exceeds the across-crop elasticity θ. I do not impose (16); I estimate its two sides separately and treat it as an overidentifying restriction on the nesting structure.[15]

6.2.4 Supply Elasticity

The own-price supply elasticity faced by mill i combines two margins of farmer response:

$$\varepsilon_i^s = \underbrace{\frac{\alpha}{\sigma}(1 - s_{i|0})}_{\substack{\text{Within-cane margin:} \\ \text{steal farmers from rival mills}}} + \underbrace{\theta\, s_{i|0}(1 - s_0)}_{\substack{\text{Cross-crop margin:} \\ \text{pull farmers from other crops into cane}}} \tag{17}$$

The within-cane margin captures substitution among mills: when mill i raises its price, it attracts farmers from competing mills, with the response governed by α/σ and proportional to $(1 - s_{i|0})$ - the share of cane farmers currently selling to other mills. The cross-crop margin captures substitution between cane and alternative crops: a higher cane price raises IV_0, pulling some plots from soybean or cotton into cane, with the response proportional to $s_{i|0}(1 - s_0)$.

The elasticity has a key comparative static for the paper's mechanism. When competitors exit following a drought, each surviving mill's within-cane share $s_{i|0}$ rises mechanically (fewer mills sharing the same farmers). This lowers ε_i^s - the mill faces a less elastic residual supply - increasing its oligopsony power. This is the market power channel of drought-induced restructuring: exit raises concentration and widens the markdown that each surviving mill can extract.

6.3 Equilibrium

Given the regulated sugar price P^s, labour wage w, fixed costs F_{op} and F_{up}, and the cane-price floor $\underline{W}^M$, a *market equilibrium* is a cane-price vector $\{W_i^M\}$, a set of mill decisions, and a set of farmer allocations such that:

[15] A proof that $\rho_0 \leq 1$ is necessary and sufficient for a valid GEV generator, via McFadden (1977), is given in the appendix.

1. **Farmers optimize.** Taking $\{W_i^M\}$ as given, each farmer chooses a crop (outer nest) and, conditional on cane, a mill (inner nest). These choices determine the residual cane supply C_i and the supply elasticity ε_i^s faced by each mill i.
2. **Mills optimize.** Taking its residual supply as given, each mill i chooses labour, cane, whether to upgrade, and whether to exit, and sets the markdown price $W_i^M \geq \underline{W}^M$.
3. **The cane market clears.** Cane supply C_d equals cane demand M_d at $\{W_i^M\}$ in each market. Mills are price-takers in the sugar and labour markets, so only the cane market clears.

Since each W_i^M enters both mill i's demand and, through $\{\delta_{i'}\}$, all farmers' shares, 1-3 constitute a fixed point in $\{W_i^M\}$. I solve it by iterating on the price vector, exploiting the monotonicity of excess cane demand.

7 Estimation

7.1 Production Function

Taking the logarithm of the production function in (3) results in equation (18). As cane is assumed to be non-substitutable and a linear function of the sugar output, it does not enter the estimable production function. The production function coefficients β need to be identified:

$$q_{it} = h(l_{it}, k_{it}, \beta) + \omega_{it}. \tag{18}$$

The "output price bias" of Klette and Griliches (1996) is taken into account because output is observed in physical units rather than as revenues. Labour is observed in physical units as well but potentially with error; rather than observing total hours worked of quality-adjusted labour l_{it}, I observe the number of employees without quality adjustment, $\tilde{l}_{it}$. Capital is measured in monetary values $\tilde{k}_{it}$. If latent differences in utilization of labour are correlated with mill productivity, this induces an "input price bias" as discussed in De Loecker et al. (2016). To address this concern, I add a function $a(\cdot)$ of wages per worker to the production function:

$$q_{it} = h(\tilde{l}_{it}, k_{it}, \beta) + a(w_{it}^L) + \omega_{it}. \tag{19}$$

Identification. In order to identify the production function, I impose timing assumptions on mills' input choices, as proposed by Olley and Pakes (1996). As was explained in Section 6.2.1, labour is assumed to be a variable and static input, whereas capital is fixed

and dynamic. Hence, labour is assumed to be chosen at time t, after the productivity shock ω_{it} is observed by the mill, while capital investment is chosen at time t–1, before the productivity shock is observed.

The usual approach in the literature is to invert the intermediate input demand function to recover the latent productivity level ω_{it}, which can be used to construct the innovation using a productivity law of motion (Levinsohn and Petrin, 2003; Ackerberg, Caves and Frazer, 2015). This approach hinges on productivity being the only latent, serially correlated input demand shifter. However, input demand varies due to markdown variation as well, as pointed out by Doraszelski and Jaumandreu (2021). This becomes a key concern in my setting given drought shocks lead to concentrated markets following exit of less productive mills, potentially altering cane price markdowns observed by surviving mills. The approach with input inversion can still be used when making additional parametric assumptions on the distribution of markdowns, as I do in my nested supply model. Alternatively, one can also impose more structure on the productivity transition process, with the benefit of allowing for fully flexible heterogeneity in markdowns.

I estimate the production function using the former approach with the benefit of allowing for endogenous exit (Olley and Pakes, 1996; Levinsohn and Petrin, 2003; Ackerberg, Caves and Frazer, 2015). I use cane as the flexible input for first stage inversion, and include cane price as an argument in the first stage regression, as explained below. I include the dynamic panel approach following Blundell and Bond (2000) as a robustness check in the Appendix.

First stage. The cane demand derived in (10) depends on cane price, sugar price, sugar recovery, and all other input prices and quantities, which are all observed. Cane demand also depends on the output elasticities of labour and capital, the markdown and on productivity, which are all latent.

$$M_{it} = m\left(P_t^s, W_{it}^M, W_{it}^L, K_{it}, \beta^L, \beta^K, \beta_{it}^M, \Omega_{it}, \mu_{it}\right) \tag{20}$$

The output elasticities β^L and β^K are assumed to be the same for all mills and time periods. Markdowns and productivity vary, however, across mills and time. Without making further assumptions on the distribution of at least one of these two variables, the scalar unobservable assumption is violated, and Ackerberg, Caves and Frazer (2015) is not identified. Input demand can vary due to productivity and/or markdowns.

However, given the nested supply model for cane, all variation in markdowns is captured by observed sugar and cane prices, and cane market shares. A similar assumption was

made in De Loecker et al. (2016) for a setting in which there is only imperfect competition downstream. Therefore, I include sugar price P^s_{it}, cane price W^M_{it}, and market shares $s_{it} \equiv (s_{j|0}, s_0)$ in the first stage regression, which is given by (21). The market shares included are total cane market share, and the inner-nest cane market share. Measurement error ε^q_{it} is assumed to be additive to the function $\phi_t(\cdot)$.

$$q_{it} = \phi_t(\tilde{l}_{it}, k_{it}, W^M_{it}, W^L_{it}, P^s_{it}, \beta^M_{it}, s_{it}) + \varepsilon^q_{it} \tag{21}$$

I estimate this model using a second-order polynomial in all inputs to purge measurement error and recover fitted $\hat{\phi}_{it}$. Productivity can then be recovered as a function of the data and estimable parameters.

$$\omega_{it} = \hat{\phi}_{it} - h(\tilde{l}_{it}, k_{it}, \beta) - a(W^L_{it}) \tag{22}$$

Second stage. I allow for endogenous exit by estimating an exit probability $\hat{p}^x_{it}$ using a probit regression of observed exit on all the variables in the input demand function. I allow for this exit probability to be endogenous to productivity, as proposed by Olley and Pakes (1996), and following the implementation of Collard-Wexler and De Loecker (2015). I also allow productivity to depend on a drought dummy, $\mathbf{D}_{it}$, to reflect changes in local market structure as in De Loecker (2013) and Braguinsky et al. (2015).

$$\omega_{it} = g(\omega_{i,t-1}, \hat{p}^x_{i,t-1}, \mathbf{D}_{it}) + \nu_{it} \tag{23}$$

The productivity innovation ν_{it} is given by the difference between productivity and its expected value from the law of motion.

$$\nu_{it} = \omega_{it} - \mathbb{E}(\omega_{it}|\omega_{it-1}) \tag{24}$$

Since labour is static and capital is predetermined, the innovation ν_{it} is orthogonal to lagged labour and current capital, giving the moment conditions

$$\mathbb{E}\left[\nu_{it}\left(\tilde{l}_{i,t-1},\, k_{it},\, w^L_{it}\right)\right] = 0 \tag{25}$$

I impose a third-order polynomial function for ω_{it-1} and a linear function for $\hat{p}^x_{i,t-1}$ and Z_{it} in the productivity law of motion, and block-bootstrap standard errors with 200 iterations.

Results. The estimated output elasticities are in Table 1 below. The estimates using the control function approach are in Panel B, and are 0.715 and 0.317 for labour and capital

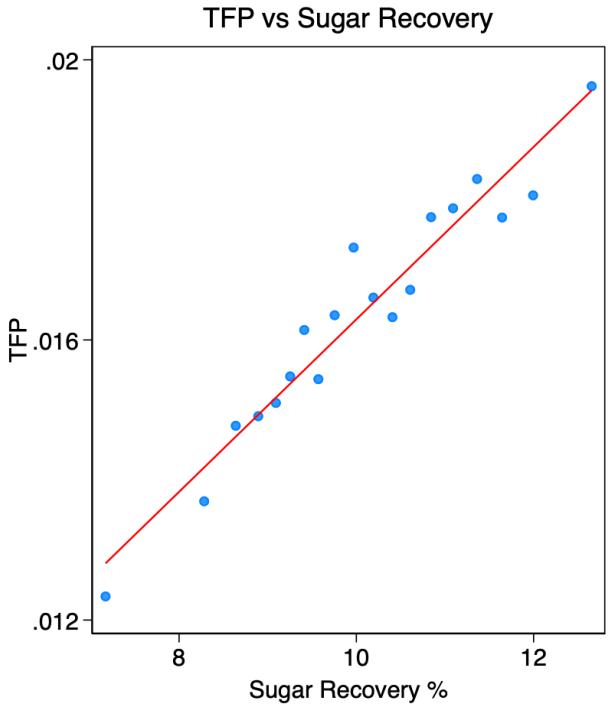
respectively. These estimates are respectively lower and higher compared to the OLS estimates in Panel A, which are 0.740 and 0.285 for labour and capital. Both specifications have a scale parameter that is just above one. Figure 10 plots estimated TFP against recovery, highlighting a strong positive relationship between the two productivity measures.

Table 1: Production Function Estimates

	A. Ordinary Least Squares		**B. Control Function**	
	Estimate	SE	Estimate	SE
Output Elasticity of Labour	0.740	0.012	0.715	0.064
Output Elasticity of Capital	0.285	0.004	0.317	0.022
Scale Parameter	1.025		1.032	
Observations	10,620		5,788	

Notes: Standard errors are block-bootstrapped with 200 iterations.

Figure 10: TFP and sugar recovery exhibit a strong positive relationship



Notes: Figure shows a binscatter plot for estimated TFP using the control function approach following Ackerberg, Caves and Frazer (2015) against sugar recovery.

7.2 Supply Elasticities and Markdowns

Results. Panel A of Table 2 shows OLS and IV estimates for α/σ, the substitution elasticity between mills. Panel B shows estimates for θ, the substitution elasticity between crops.

As expected, mills are closer substitutes for one another than cane is for competing crops, suggesting substantial crop switching costs. Note that the cross-crop elasticity estimate is in line with values from the agricultural trade literature, which typically finds substitution elasticities across commodities ranging between 1.5 and 2.5 (Bergquist et al., 2022; Costinot, Donaldson and Smith, 2016; Dominguez-Iino, 2026; Sotelo, 2020). Together, the two elasticities pin down the dissimilarity $\rho_0 = \frac{\theta\sigma}{\alpha}$ at 0.34, which lies between 0 and 1, thereby satisfying the consistency of the two nests.

Table 2: Supply-side Substitution Elasticities

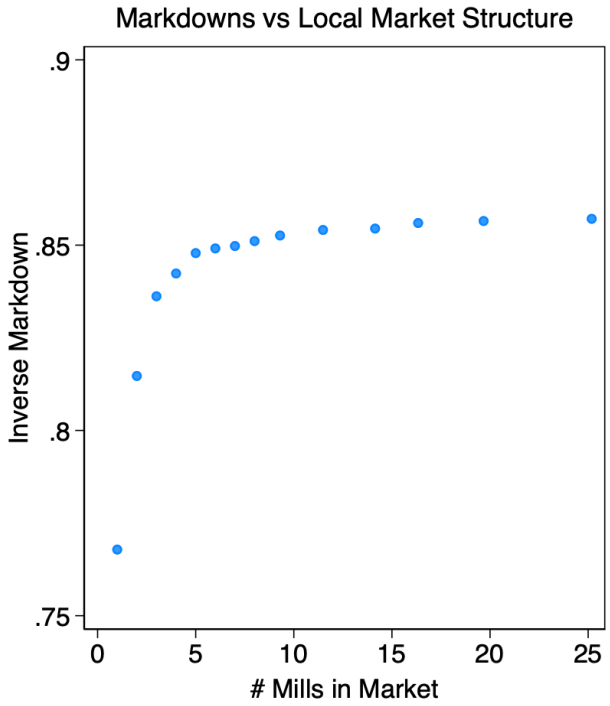
	A. Cross Mill Substitution		**B. Cross Crop Substitution**	
	OLS	IV	OLS	IV
α/σ	1.067***	6.204***		
	(0.176)	(1.217)		
θ			0.450***	2.120***
			(0.061)	(0.340)
District FE	✓	✓	✓	✓
State-Year FE	✓	✓	✓	✓
Observations	8,618	8,618	21,178	21,178
KP Wald F-statistic		152.1		42.5

Markdowns. Estimated markdown is 1.25 at the average sugar mill, and 1.19 at the median sugar mill. This implies that farmers selling to the average mill receive around 80% of cane's marginal revenue product. This number is largely comparable to figures for other agricultural goods in the literature as can be seen in Table 3. Figure 11 shows how inverse cane markdowns vary as a function of local market structure. Farmers in near-monopsony receive 72% of cane's marginal revenue product, while this figure exceeds 85% in markets with more than 15 mills.

Table 3: Average Markdowns in Agricultural Goods

Paper	Markdown	Input
Rubens (2023)	2.90	Tobacco Leaf
Zavala (2026)	1.64	Various Crops
Dominguez-Iino (2026)	1.35	Cattle
This Paper	**1.25**	**Sugarcane**
Avignon and Guigue (2025)	1.19	Raw Milk
Dominguez-Iino (2026)	1.16	Maize, Soybean
Crespi and Sexton (2005)	1.10	Cattle, Potato, Rice

Figure 11: Farmers in competitive markets receive a larger share of cane's MRP



Notes: Figure shows a binscatter plot for inverse cane markdowns against local market structure – defined by the number of mills located in the district.

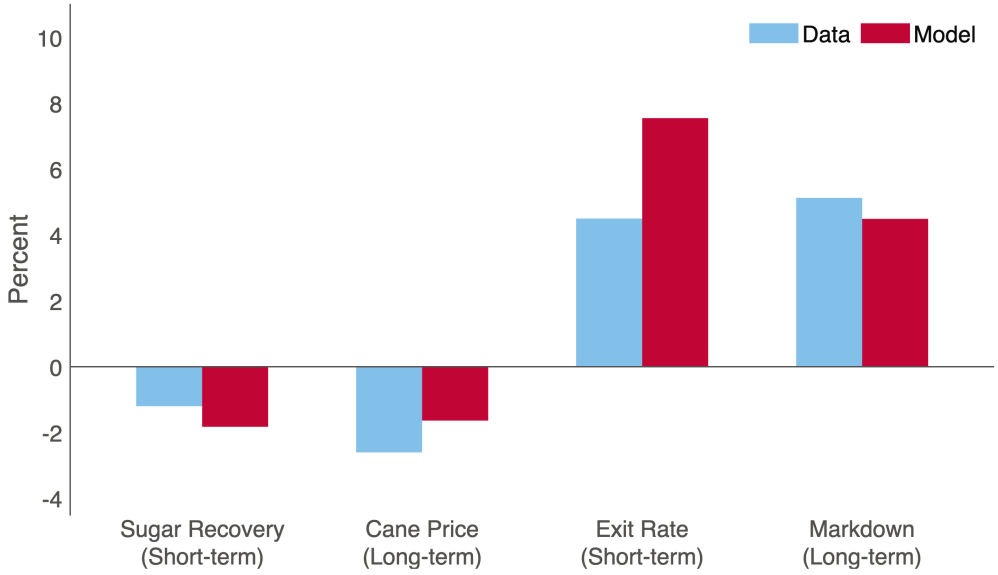
7.3 Fixed Costs

7.3.1 Fixed Costs of Operation

7.3.2 Fixed Costs of Upgrading

7.4 Summary and Model Fit

Figure 12: Comparison of model and data estimaes

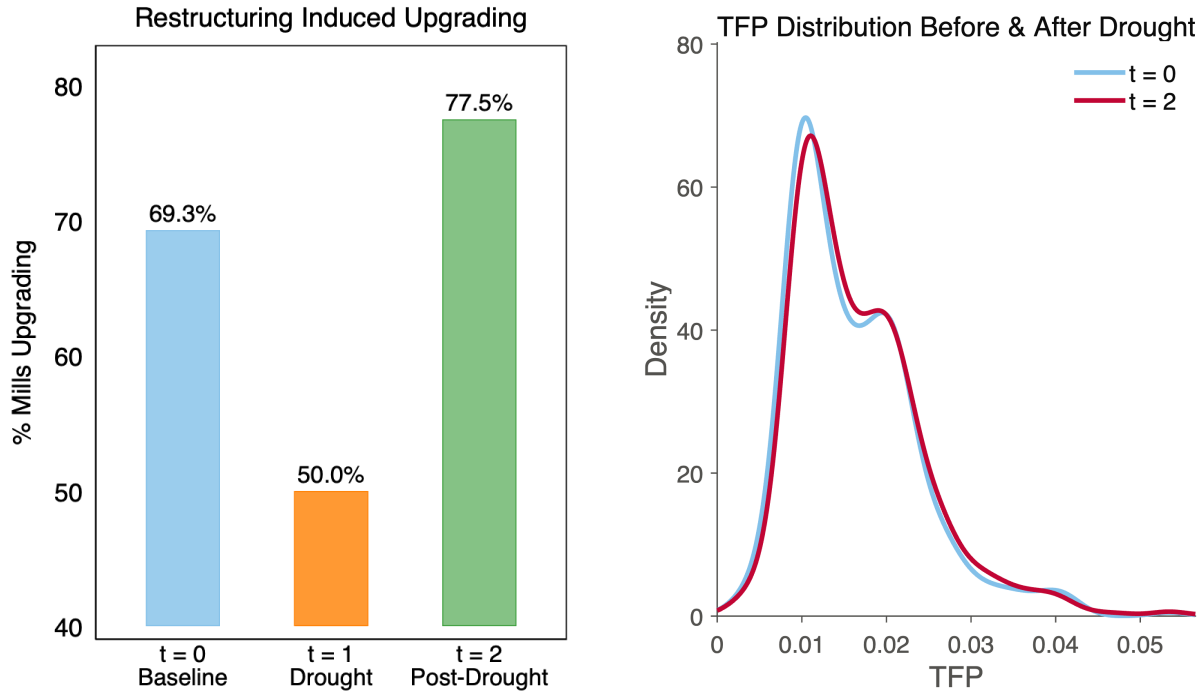


Notes: This figure compares model-predicted and empirical estimates of short- and long-run responses to drought shocks. The figure presents four key outcomes: (1) Sugar recovery shows the short-run percentage decline in mill-level recovery; (2) Cane price shows the long-run percentage decline in farm-gate cane prices; (3) Exit rate shows the short-run percentage decline in number of active mills; and (4) Cane markdown shows the percentage increase in long-run markdowns imposed by mills on cane prices.

8 Aggregate Effects

8.1 Productivity Decomposition

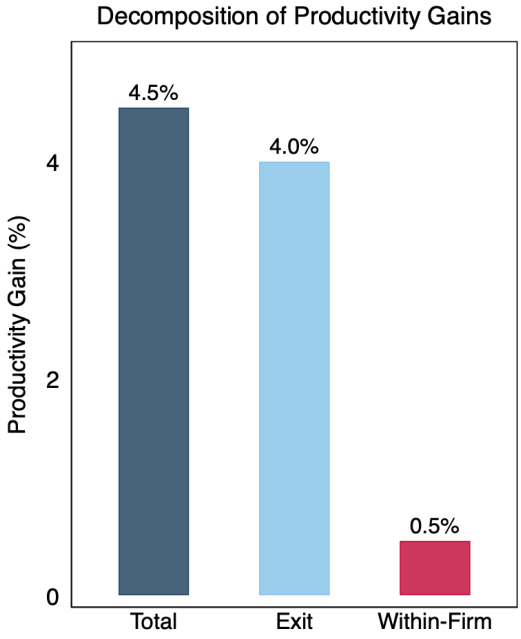
Figure 13: Market restructuring induced upgrading and productivity gains



Panel A: Restructuring-induced Upgrading **Panel B:** TFP Distribution

Notes: This figure simulates a drought shock within the estimated model by simulating a drought-induced reduction in crop yields that matches event-study estimates from Section 5. Panel A tracks the share of sugar mills in impacted markets that engage in cane management upgrading in the short- and long-run. Panel B illustrates the aggregate productivity gains by presenting kernel density plots of the TFP distribution before and after the shock, highlighting shifts toward higher productivity due to two adjustments: within firm gains and extensive-margin effects due to exit.

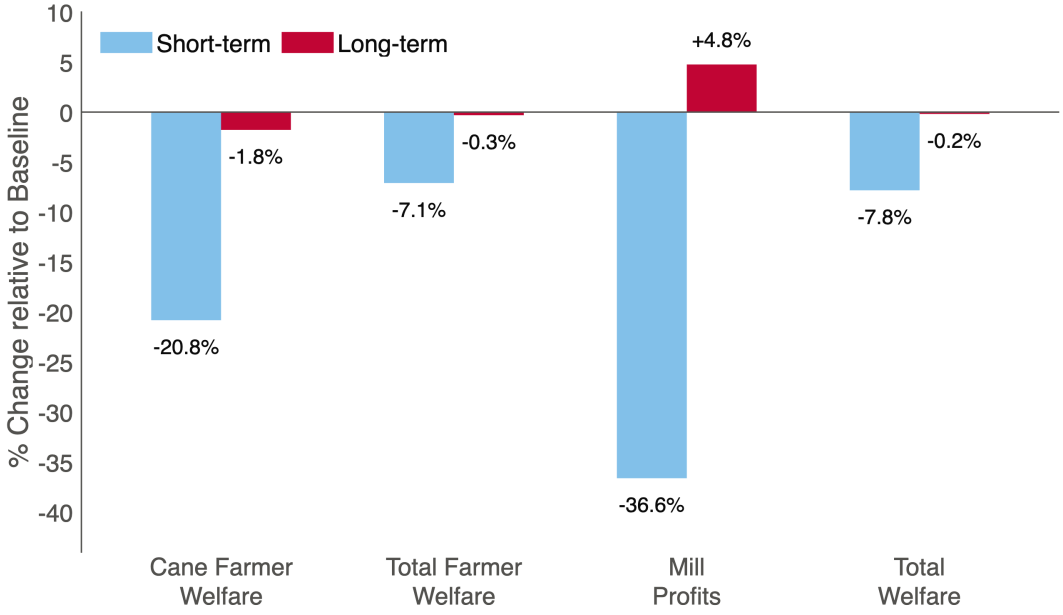
Figure 14: Aggregate mill productivity rises by 4.5% in the long-term



Notes: This figure decomposes the total productivity increase into two mutually exclusive components: selection and within firm gains, underscoring the distinct mechanisms through which aggregate productivity is enhanced in response to drought-induced market restructuring.

8.2 Welfare Decomposition

Figure 15: Cane farmer welfare declines by 1.8% in the long-term



9 Conclusion

This paper shows that climate change impacts farmers not only by lowering crop yields, but also by restructuring the markets in which they sell. In the Indian sugar industry, extreme droughts cause the least productive mills to exit and their farmers to reallocate to nearby survivors. This reallocation is not merely compositional: by expanding the volume base over which quality improvements apply, it raises survivors' return to upgrading and triggers within-firm productivity growth. But the same concentration that funds upgrading also widens mills' buyer power: farm-gate cane prices do not rise, so the productivity gains are captured by mills rather than shared with the millions of farmers who supply the industry. Climate-induced adaptation and exploitation are thus complements, driven by a single change in local market structure.

These findings carry a broader lesson for how we think about climate change and agricultural markets. The welfare consequences of a climate shock for smallholders depend not only on its direct effect on yields, but on how it reshapes the market power of the intermediaries they depend on – a channel largely absent from existing work, which focuses on direct weather-to-yield effects. Where intermediaries adapt, aggregate productivity can rise even as farmers are left worse off, so the efficiency gains from climate-induced restructuring and its distributional costs are two faces of the same mechanism. This poses a genuine policy trade-off: interventions that prevent exit forgo both the reallocation gains and the within-firm upgrading that reallocation enables, while doing nothing leaves farmers exposed to widening markdowns. In future work, I plan to evaluate policies that target the distributional margin directly while preserving the upgrading that concentration sets in motion.

As climate shocks become more frequent, understanding how they redraw the boundaries of market power, and not only the productivity of the land, becomes central to protecting smallholders who remain at the base of global agricultural supply chains.

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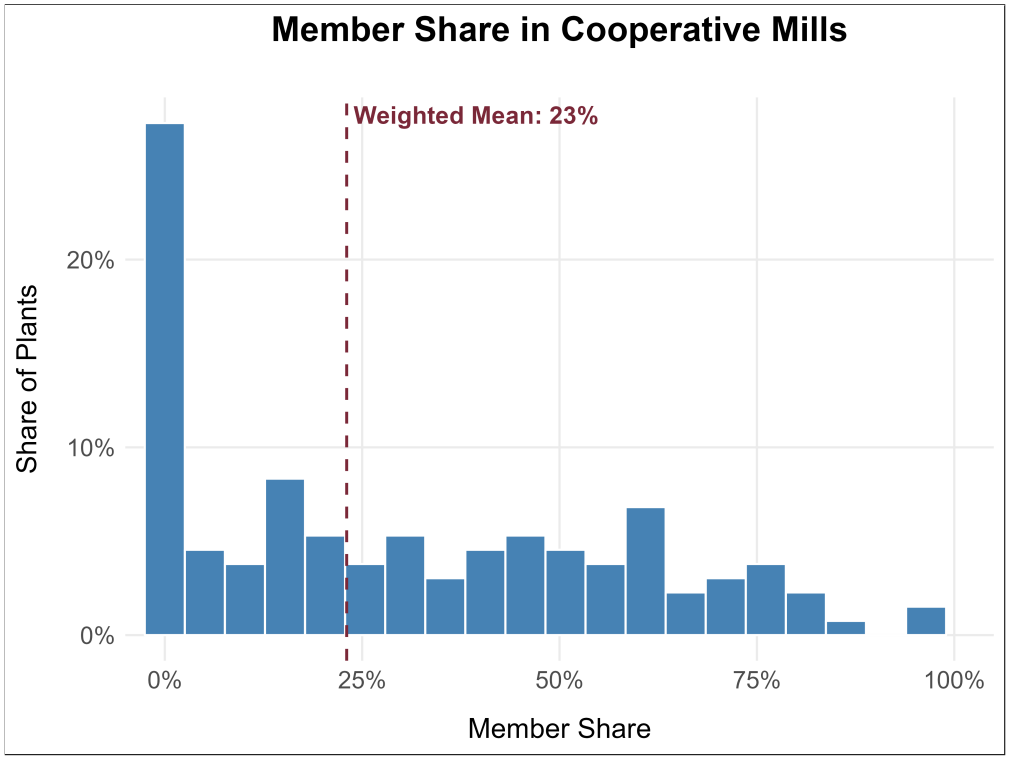
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A Appendix

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A.1 Figures

Figure A.1: Share of Members in Cooperative Mills



Notes: This figure plots the share of member farmers in cooperative mills across Maharashtra in 2025-26. *Source:* Farmer-Mill Transactions and Plant-level Records from the Maharashtra Sugar Commissionerate.

Figure A.2: Industrial Water Requirement

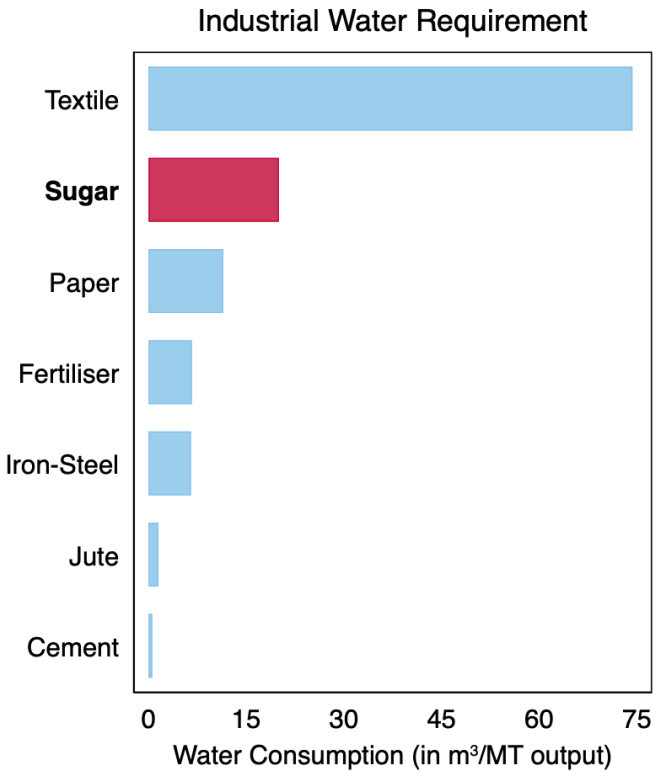
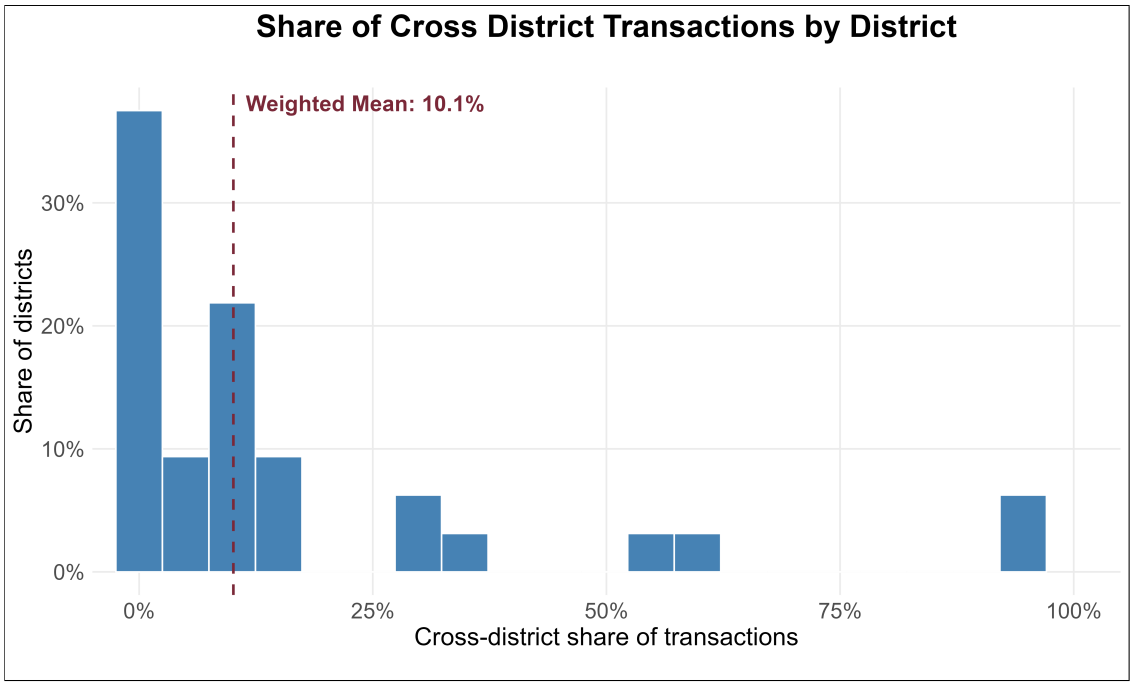


Figure A.3: Newspaper Headlines on Drought-induced Mill Closures



Figure A.4: Share of Cross District Farmer-Mill Transactions



Notes: This figure plots the share of farmer-mill transactions that take place across district boundaries. *Source:* Farmer-Mill Transactions and Plant-level Records from the Maharashtra Sugar Commissionerate.

Figure A.5: Monthly Sugar Release Allocations

Sl. No.	Plant Code	Plant Name	Notional Stock as on 31.07.2024	Quota calculated as per Notional Stock
(1)	(2)	(3)	(4)	(5)
	State:	**Punjab**		
383	101	GURDASPUR	11466	2371
384	201	BATALA	9126	1887
385	301	FAZILKA	3111	643
386	401	MORINDA	15514	3208
387	501	BHOGPUR	17270	3571

Notes: This figure shows monthly sugar release allocations outlined by the Indian Directorate of Sugar for sugar mills in Punjab for August 2024. *Source:* Indian Directorate of Sugar.

A.2 Tables

A.3 Theoretical Appendix